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SIRKM 2025

**SEMINAR ON INFORMATION RETRIEVAL
AND KNOWLEDGE MANAGEMENT 2025**

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SIRKM²⁰²⁵

SEMINAR ON INFORMATION RETRIEVAL
AND KNOWLEDGE MANAGEMENT 2025

E-Proceedings of

Seminar on Information Retrieval and Knowledge Management 2025 (SIRKM'25)

Editors:

Dr. Azilawati Azizan
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FOREWORD

It is my great pleasure to welcome all distinguished participants, presenters, and guests to the Seminar on Information Retrieval and Knowledge Management 2025 (SIRKM'25). This biennial seminar holds a special place in our community as a platform that brings together scholars whose work continues to shape the future of information retrieval, knowledge management, and intelligent digital ecosystems.

This year's SIRKM'25 is jointly organised by Universiti Teknologi MARA (UiTM), the Persatuan Capaian Maklumat dan Pengurusan Pengetahuan Malaysia (PECAMP). This collaboration reflects our shared commitment to strengthening research networks, fostering interdisciplinary dialogue, and elevating Malaysia's role within the regional and global knowledge landscape.

The seminar welcomes extended abstracts across a diverse range of themes, spanning Information Retrieval, Knowledge Management, Artificial Intelligence and Data Science, Information Visualisation, and emerging areas such as Cyber Security, IoT, and Industrial Computing. I am delighted to note that we have received 20 extended abstracts, representing both local and international contributions. These works demonstrate the vibrancy and relevance of our research community, and selected papers will be considered for journal publication, further amplifying their scholarly impact.

This year, SIRKM'25 also features a dedicated Workshop on Generative AI for Research, reflecting the rapid transformation brought about by frontier technologies. As AI continues to reshape how knowledge is created, accessed, and applied, PECAMP remains committed to nurturing responsible, meaningful and future-ready innovation among our researchers and professionals.

A highlight of the seminar is the SIRKM'25 Best Award, which recognises excellence in research quality, relevance, and contribution. I congratulate all contributors for their dedication and encourage you to continue pushing boundaries in your respective fields.

On behalf of PECAMP, I extend my heartfelt appreciation to our organising partners, programme committee, reviewers, speakers, and volunteers whose unwavering efforts have made SIRKM'25 possible. I hope that the interactions through SIRKM'25 will inspire new collaborations, spark innovative ideas, and strengthen our collective mission to advance information science and knowledge management in Malaysia and beyond.

Wishing everyone a productive, insightful, and inspiring seminar.

Prof. Ts. Dr. Nurfadhlin Mohd Sharef
President
Persatuan Capaian Maklumat dan Pengurusan Pengetahuan (PECAMP)

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A DATA SCIENCE–DRIVEN ECOLOGY SIMULATOR FRAMEWORK FOR PRECISION BIODIVERSITY AND FOREST STAND MANAGEMENT

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ABSTRACT Tropical rainforests such as those in Malaysia represent some of the most biodiverse yet vulnerable ecosystems on Earth. Effective forest stand management requires the ability to analyze and simulate biodiversity dynamics across multiple species groups, environmental factors, and time scales. However, existing forest modeling systems such as SORTIE-ND, FORMIND, LANDIS-II, and 3-PG are predominantly designed for temperate ecosystems and lack flexibility to represent the complex, mixed-dipterocarp forests typical of Southeast Asia. This study introduces a data science–driven Ecology Simulator Framework developed for precision biodiversity (precision bioD) and sustainable forest stand management. The framework integrates heterogeneous ecological datasets through an Integrated Data Lake and links them with two core modules: Biodiversity Dynamics Assessment (BDA) for monitoring and analyzing forest composition, and Biodiversity Dynamics Simulation (BDS) for modeling stand trajectories, harvesting scenarios, and biodiversity implications. The paper’s main contribution lies in establishing a systematic data science methodology for ecological simulation—an area with limited existing frameworks. The methodology emphasizes iterative development, expert validation, and decision-oriented analytics. Validation with forestry and ecology experts confirmed the framework’s technical accuracy, usability, and policy relevance. The results demonstrate its potential to transform long-term ecological data into actionable insights for adaptive management, biodiversity conservation, and sustainable forest governance.

Keywords: Ecology Simulator, Biodiversity Dynamics, Precision Biodiversity, Data Science Framework, Decision Support System

1. INTRODUCTION

Tropical rainforests such as those in Malaysia hold extensive ecological data, yet their potential remains underexploited. Despite long-term datasets from sites like the Pasoh Forest Reserve [1], there is no structured data science framework to integrate and translate these data into predictive insights for biodiversity and forest management. Existing models such as SORTIE-ND [2], FORMIND [3], LANDIS-II [4], and 3-PG [5] are limited by static parameterization and low adaptability to complex tropical ecosystems. This gap underscores the need for a data science–driven ecology simulator that unifies biodiversity analytics and simulation for adaptive, evidence-based forest decision support [6].

To bridge these gaps, this research introduces a data science–driven Ecology Simulator Framework specifically designed for precision biodiversity management and forest stand decision support in tropical ecosystems. The framework integrates analytics and simulation through two synergistic components:

1. Biodiversity Dynamics Assessment (BDA): focuses on quantifying, analyzing, and visualizing temporal changes in forest composition, structure, and diversity using long-term ecological datasets.
2. Biodiversity Dynamics Simulation (BDS): models and forecasts stand-level trajectories and biodiversity outcomes under varying environmental, silvicultural, and policy-driven management scenarios.

Together, these components form a Forest Stand Management Decision Support System (FSM-DSS) that generates adaptive, data-informed insights on species group behavior (e.g., dipterocarps, pioneers, palms), ecosystem resilience, carbon dynamics, and economic trade-offs.

Beyond the system itself, the core contribution of this study lies in its methodological innovation. While

numerous models simulate forest dynamics, there is no established data science methodology that systematically integrates biodiversity dynamics analytics with ecological simulation within a DSS architecture. This research therefore contributes a new methodological foundation for precision biodiversity, a discipline that applies advanced data analytics, model integration, and iterative validation to enhance understanding and management of ecosystem complexity. The proposed iterative design, encompassing six refinement cycles, establishes a reproducible process for data-driven ecological modeling, biodiversity intelligence extraction, and scenario-based forest decision support.

By uniting principles from data science, ecology, and sustainability analytics, the Ecology Simulator Framework aims to advance both the scientific understanding of tropical forest dynamics and the methodological rigor of ecological decision systems, paving the way for next-generation digital twins for forest ecosystems and biodiversity conservation.

2. ECOLOGY SIMULATOR FRAMEWORK

The Ecology Simulator Framework (Figure 1) integrates data science methodologies with ecological modeling to support biodiversity dynamics analysis and forest stand management decision-making. It consists of three main modules namely Data Pre-processing, BDA, and BDS which are interconnected through an Integrated Data Lake and a Forest Management Knowledge base. The Data Pre-processing module consolidates heterogeneous data from flora, environmental, and forest sources through selection, integration, and transformation, ensuring quality and compatibility for downstream analysis. The BDA module enables monitoring and analysis of biodiversity indicators, generating overviews, trends, and query-based insights on species composition, structure, and diversity changes across censuses. The BDS module extends these insights through simulation, employing DBH estimation and forest knowledge to model stand dynamics, harvesting scenarios, and long-term management implications.

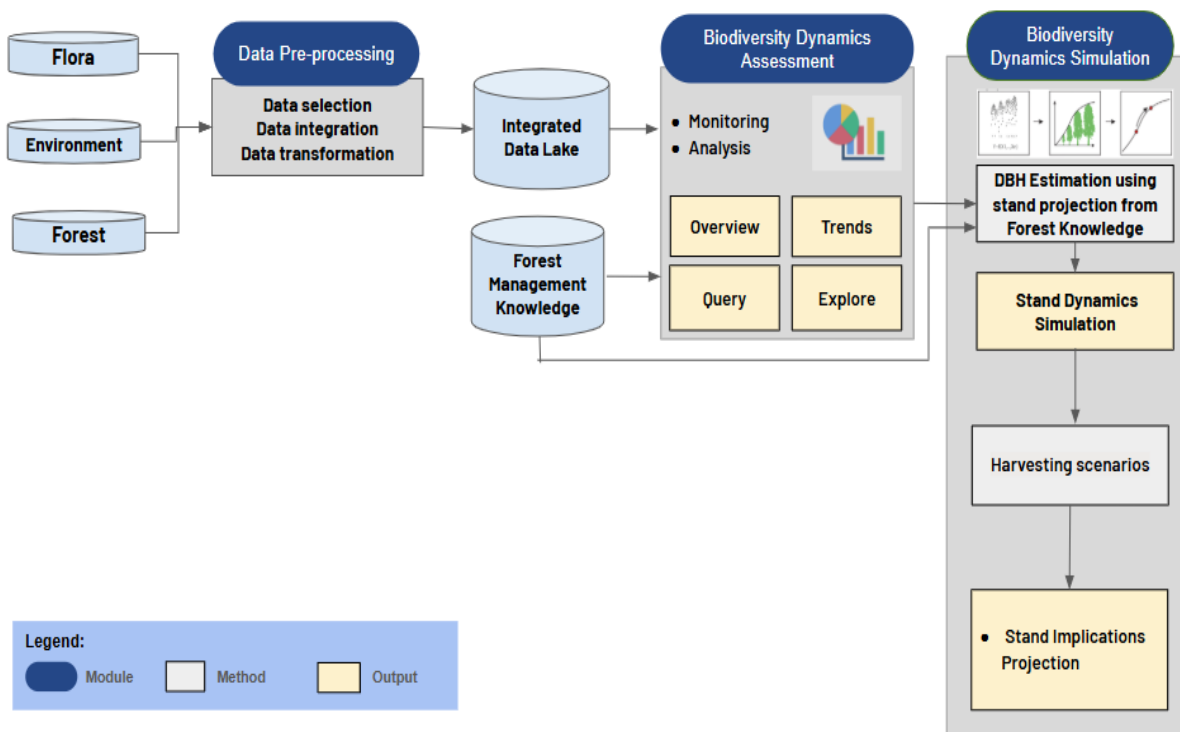


Figure 1. Framework of Ecology Simulator

3. RESULTS AND FINDINGS

To ensure the robustness, usability, and scientific validity of the proposed Ecology Simulator Framework, a multi-phase expert validation process was conducted. Table 1 summarizes the objectives, phases, and outcomes of each validation stage, highlighting how expert feedback contributed to the iterative refinement of the Ecology Simulator Framework.

Table 1. Activities in Ecology Simulator system development

Phase	Objectives	Key Findings
Preliminary Evaluation	- Assess the initial usability and practicality of the dashboard in monitoring biodiversity dynamics. - Gather user feedback to refine the design and improve overall system functionality.	Usability assessment report produced, detailing system strengths, limitations, and overall user experience rating. Feedback informed interface redesign and data visualization improvements.
BDA Design Finalization	- Finalize the design of the BDA module. - Synchronize and standardize the definitions of key terms, metrics, and formulas.	Final BDA module design endorsed by experts, validated for clarity, metric consistency, and usability across biodiversity datasets.
Design Verification – BDA	- Verify the progress and requirements of the BDA module. - Validate specifications for Biodiversity Dynamics Prediction (BDP) and BDS components.	Expert panel reviewed the BDA progress and validated the BDP and BDS requirement documents, confirming alignment with intended analytical and simulation outputs.
Design Verification – BDS	- Finalize the design of the BDS module. - Validate integration between BDA and BDS reports for decision support readiness.	Approved the simulation interface, visualization components, and reporting templates for BDA–BDS integration within the DSS environment.
First Evaluation	- Evaluate the perceived usefulness, relevance, and insightfulness of the dashboard in supporting biodiversity monitoring and decision-making. - Identify dashboard features that enhance or limit expert interpretation.	Comprehensive expert validation report compiled, highlighting effective visualization features, usability gaps, and prioritized improvement strategies for subsequent iteration.

4. CONCLUSION

This paper presents the Ecology Simulator Framework, a data science–driven approach for precision biodiversity (precision bioD) and sustainable forest stand management. Integrating multi-decadal ecological data through an Integrated Data Lake, it unites Biodiversity Dynamics Assessment (BDA) and Biodiversity Dynamics Simulation (BDS) for analysis and predictive modeling. Expert validation confirmed its technical robustness and relevance to species-rich tropical forests. The framework bridges data science and biodiversity analytics, offering a new methodological foundation for ecological simulation. Future development will extend toward remote sensing and climate integration, evolving into an intelligent digital twin ecosystem for tropical forest governance.

ACKNOWLEDGEMENT

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VISUAL ANALYTICS FOR UNDERSTANDING BANK CUSTOMERS' PERCEPTIONS IN MALAYSIA

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ABSTRACT This study explores how data visualization serves as an analytical tool to examine public sentiment toward three anonymized Malaysian banks—Bank A, Bank B, and Bank C—based on social media data. The research focuses on how visual analytics can reveal trends, sentiments, and relationships between customers and banks. Using Twitter data, the study applies a structured data analysis process that includes data cleaning and visualization to interpret customer opinions and satisfaction levels. The findings show that visualization simplifies complex data and reveals key insights about trust, service quality, and digital engagement. The findings prove that Bank A demonstrates the most positive public perception, Bank B shows moderate satisfaction with mixed reviews, and Bank C reflects ethical strength but lower digital engagement. The study concludes that visualization enhances the understanding of customer perception and supports decision-making to improve banking services.

Keywords: Visual Analytics, Customer Perception, Sentiment Analysis, Banking Industry, Data Visualization

1. INTRODUCTION

As banking services increasingly rely on digital platforms, understanding how customers perceive banks has become essential in maintaining trust and satisfaction. Traditional methods such as surveys provide limited insights, whereas visual analytics enables organizations to explore real-time customer opinions dynamically [1]. Visualization transforms large volumes of unstructured data into meaningful visual patterns, allowing analysts to identify trends and sentiments intuitively [2,3].

This study investigates customer perceptions of three major anonymized Malaysian banks—Bank A, Bank B, and Bank C by applying visualization techniques to social media data. The focus is not on algorithmic modeling but on how visual representation enhances understanding of customer sentiment. Visualization functions as the core analytical tool that connects data interpretation with business insight, helping financial institutions identify strengths, weaknesses, and areas for improvement in customer engagement [4, 5].

While many studies have applied sentiment analysis to assess customer opinions in the financial sector, most rely on machine-learning or lexicon-based models that prioritize prediction accuracy over interpretive understanding. These models often generate numerical outputs without translating them into meaningful insights for decision-makers. Mohanarajesh [6] highlighted that deep learning approaches in financial sentiment analysis remain limited by transparency and explainability, while Peterson [5] emphasized the importance of human-centered visualization to bridge analytical complexity and cognitive interpretation. Despite the growing adoption of social media analytics, few studies have explored visualization as a central analytical approach to interpret customer perception [3]. Therefore, there remains a gap in exploring visual analytics as a communicative and diagnostic tool—not merely as a supplement to quantitative sentiment analysis. This study addresses that gap by demonstrating how visualization can complement statistical sentiment measures to deliver a clearer and more actionable understanding of customer perceptions toward major Malaysian banks [3],[5],[7].

The remainder of this paper is structured as follows: Section 2 describes the methodology and data sources used; Section 3 presents the visualization results and analysis; and Section 4 concludes the study with key findings and recommendations for future work.

2. METHODS AND DATA DESCRIPTION

The dataset was obtained from Twitter posts mentioning Malaysian banking institutions. Tweets were collected using the Selenium WebDriver, which automated searches for keywords and hashtags related to major banks. Initially, tweets referencing all major Malaysian banks were gathered, and after screening, only the top three—Bank A, Bank B, and Bank C—were retained for detailed analysis, as these institutions generated the highest number of customer-related comments and engagement activity. The final dataset comprised approximately 4,500 tweets, with fewer than 1,500 entries per bank. Each record represented a single tweet and included the following attributes: Tweet_ID, Timestamp, Bank_Name, Tweet_Text, Language, and Sentiment_Label (positive, neutral, or negative).

Prior to analysis, several data-cleaning steps were performed to ensure accuracy and consistency. Non-related mentions, retweets, and promotional advertisements were removed to retain only genuine customer opinions. Duplicate entries, emojis, and irrelevant symbols were eliminated, and Malay-language tweets were translated into English to standardize textual content. Additionally, the text was converted to lowercase. Punctuation, URLs, and stop words were removed for clarity. After cleaning, each tweet was categorized into one of three sentiment classes—positive, neutral, or negative, using a hybrid approach that combined keyword-based polarity scoring with manual validation. Positive tweets conveyed satisfaction or praise (e.g., “Bank B’s mobile app is convenient and reliable.”), neutral tweets reflected factual or non-opinionated statements (e.g., “Opened a Bank A account this morning.”), and negative tweets highlighted minor dissatisfaction (e.g., “Bank C’s app occasionally takes too long to load.”). These refined attributes provided a reliable foundation for the exploratory and descriptive visualization analysis conducted in this study.

Visualization dashboards were developed using Microsoft Excel and Python’s Matplotlib library to summarize and explore sentiment data. Bar charts, radar plots, and trend visualizations were used to illustrate sentiment polarity and comparative performance among the three banks.

3. RESULT AND DISCUSSION

This section presents the findings on customer sentiment toward Malaysian banks on Twitter, examining patterns, engagement levels, and temporal trends through exploratory data analysis (EDA) and visualizations, while highlighting the impact of digital services, customer experience, and ethical reputation on public perception.

3.1 Exploratory Data Analysis (EDA) and Overview of Customer Sentiment

An exploratory data analysis (EDA) was conducted to understand the overall structure and distribution of the Twitter dataset before detailed interpretation. EDA revealed that Bank A, Bank B, and Bank C dominate Malaysian banking discussions on Twitter, with varying degrees of engagement and sentiment balance. Descriptive statistics showed that positive sentiments form the majority across all banks, reinforcing public trust in the financial sector. Approximately 79.4% of respondents agreed that social media accurately reflects their banking experiences.

Visualization using the radar chart (Figure 1) highlights that all three banks maintain a largely positive sentiment base, though engagement intensity differs. The temporal trend visualization (Figure 2) further illustrates monthly fluctuations in sentiment, showing that all three banks experienced positive growth in perception over time, with Bank A displaying the steepest upward trajectory.

3.2 Bank A: Strong Brand Trust and Digital Leadership

Bank A's results revealed consistently high positive sentiment (around 60%), supported by gradual and steady engagement growth over time. Visualization confirmed this by highlighting frequent mentions related to mobile-app usability, customer service efficiency, and digital reliability. Customers commonly praised Bank A's responsive online platform and seamless transactions, while occasional negative feedback referred to short maintenance periods or temporary access delays.

The sentiment trend in Figure 2 shows a moderate upward trajectory, rising from approximately 55 % to 63% over the six-month period. This pattern indicates progressive improvement in customer confidence, with noticeable increases following digital-service updates and promotional activities, reinforcing Bank A's reputation for technological leadership and service dependability.

3.3 Bank B: Moderate Satisfaction with Service Concerns

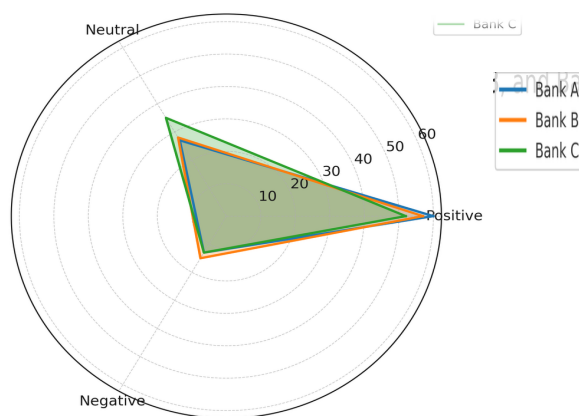


Figure 1. Comparative Sentiment Radar Chart for Bank A, Bank B, and Bank C.

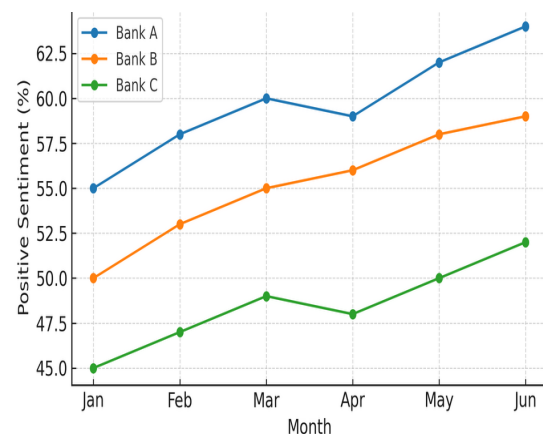


Figure 2. Monthly Sentiment Trend for Bank A, Bank B, and Bank C.

Bank B's sentiment profile showed a moderately positive distribution, with approximately 57% positive, 28% neutral, and 15% negative mentions. Visualization indicated that customers appreciate Bank B's online accessibility and user-friendly services, though some expressed concerns regarding response delays and intermittent service quality.

As shown in Figure 2, Bank B's trend line demonstrates a gradual upward movement from roughly 50% to 59%, suggesting steady but moderate improvement in customer satisfaction. Small dips were observed during periods of system maintenance or service interruptions, followed by recoveries aligned with marketing campaigns and product launches. This trend suggests that while Bank B is strengthening its customer relationship, maintaining consistent communication and reliable digital performance remains critical for sustaining long-term trust.

3.4 Bank C: Ethical Reputation but Low Digital Appeal

As shown in Figure 1, Bank C recorded the lowest tweet frequency among the three banks but exhibited a higher proportion of neutral sentiment (around 35%), indicating a more reserved and less emotionally expressive online community. Visualization highlights that customers appreciate Bank C's ethical reputation and Shariah-compliant banking principles, reflecting trust and alignment with its institutional values.

However, discussions also revealed limited digital innovation and slower service response, which may contribute to its relatively lower engagement levels. The trend chart in Figure 2 shows a gradual increase in positive sentiment, rising from approximately 45 % to 52 % over the observed period. This steady but modest improvement suggests that while ethical credibility maintains customer trust, enhancing the digital experience and mobile usability could help Bank C convert neutral perceptions into stronger positive engagement.

4. COMPARATIVE SUMMARY

The combined visualizations reveal a clear sentiment hierarchy among the three banks. Bank A leads in positive perception and digital engagement, Bank B maintains moderate satisfaction but fluctuates with service events, and Bank C upholds ethical credibility yet trails in digital competitiveness. These insights reinforce the role of visual analytics as a diagnostic approach to understanding public perception and guiding data-driven service improvements.

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TECHNIQUES FOR IDENTIFYING NOISY SOCIAL MEDIA TEXT PRIOR TO NORMALIZATION: A SCOPING REVIEW

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ABSTRACT The presence of noisy text in social-media data poses major challenges for natural language processing (NLP) applications. Before normalization, noisy text must be correctly identified to prevent error propagation throughout the pipeline. This scoping review maps the techniques used to detect and classify noisy text prior to normalization. Following Arksey and O'Malley's framework and PRISMA-ScR guidance, searches were conducted in Scopus, Web of Science, IEEE Xplore, ACM Digital Library, and ScienceDirect for papers published between 2020 and 2025. From 770 records, 35 studies were selected. Results reveal that explicit noisy-text identification is often neglected, with many studies embedding it inside normalization. Techniques found include rule-based heuristics, edit-distance and dictionary methods, frequency-based detection, classical machine-learning models, deep neural networks, and hybrid architecture. Common noise categories are misspellings, abbreviations, phonetic variants, slang, character elongation, and code-mixing. Persistent issues include limited annotated corpora, inconsistent taxonomies, and English-centric resources. The study highlights the need for transparent detection stages and multilingual benchmarks, particularly for Malay-English social-media contexts.

Keywords: noisy text; text normalization; social media; natural language processing; scoping review

1. INTRODUCTION

Social-media text is linguistically diverse and highly informal. Misspellings, abbreviations, phonetic spellings, and code-mixing frequently appear, degrading NLP model accuracy [1]. Although normalization aims to convert such variants into standard forms, many studies do not clearly isolate the stage of noisy-text identification [2]. Without explicit identification, it becomes difficult to compare performance across models or reproduce datasets.

This review addresses three objectives: (i) to map the techniques used for identifying noisy text prior to normalization; (ii) to synthesize the most common noisy-text categories; and (iii) to outline methodological and resource gaps, particularly in low-resource languages such as Malay [3, 4].

2. MATERIALS AND METHODS

This scoping review followed a transparent, staged process to map techniques for identifying noisy social-media text prior to normalization. The protocol combined Arksey and O'Malley's framework with JBI and PRISMA-ScR guidance [5]. Searches were conducted across major digital libraries (2020–2025), records were screened against predefined criteria, data were charted with a standardized form, and findings were synthesized thematically into methodological five approach categories (rule-based, statistical, machine learning, deep learning, hybrid). After deduplication and screening, 35 studies were included for analysis.

2.1. Review Design

The review implemented five stages: (i) defining research questions; (ii) identifying and collecting studies; (iii) eligibility screening (title/abstract, then full text); (iv) data charting using a structured template; and (v) collating, summarizing, and reporting results. The approach emphasized reproducibility and clarity of reporting.

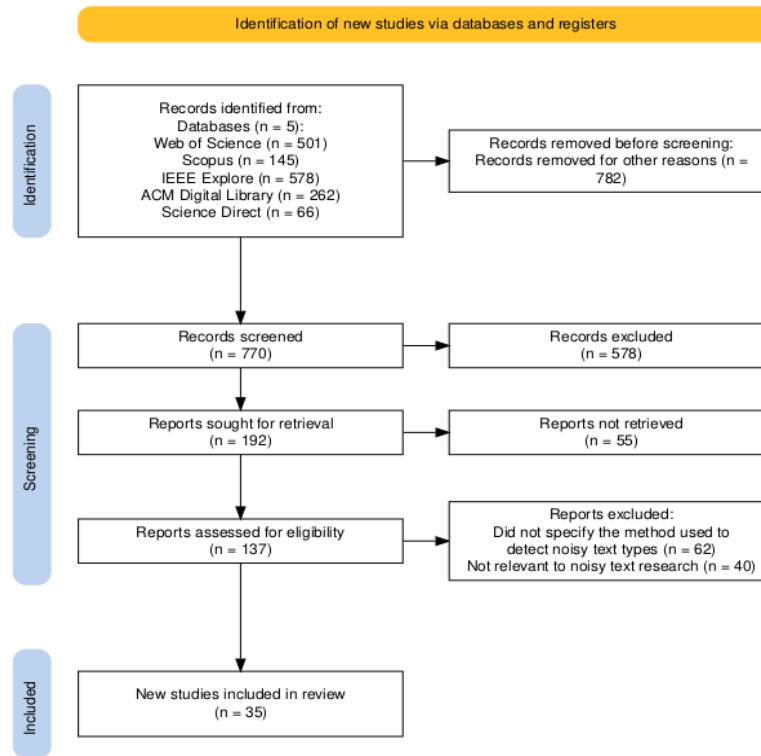


Figure 2. Scoping Review Methodology

2.2. Data Sources and Search Strategy

Five databases were searched, Web of Science, Scopus, IEEE Xplore, ACM Digital Library, and ScienceDirect, using tailored Boolean strings and controlled terms related to “noisy text,” “informal text,” “normalization,” and “social media.” The time window was 2020 - 2025 to capture current practice. Search strings were adjusted per index syntax and documented to support replication.

Table 2. Inclusion and Exclusion Criteria

Criteria	Inclusion	Exclusion
Publication Years	2020 to 2025	Published before 2020
Document Type	Articles, Conference Proceedings	Books, Chapters in Books, Book Series, Review Articles
Approach/Focus	Text normalizes, noise text, noisy text classification, error spelling correction, sentiment analysis	Studies unrelated to noisy text

2.3 Eligibility Criteria and Study Selection

Inclusion criteria targeted studies that described or enabled explicit identification/classification of noisy text in the context of normalization pipelines, focusing on social-media and comparable user-generated content. Exclusion criteria omitted purely theoretical pieces without empirical procedures, non-text modalities, and works that addressed only downstream tasks without any identifiable detection stage. Screening proceeded from titles/abstracts to full-text review.

2.4 Data Charting and Variables

A standardized charting form was used to extract relevant study details, including language or domain, corpus type, noisy-text categories (e.g., misspellings, abbreviations, phonetic variants, elongation, slang, code-mixing, and out-of-vocabulary tokens), identification technique, normalization approach, evaluation measures, and stated limitations.

2.5 Synthesis and Grouping of Techniques

Extracted methods were grouped into five major types of techniques: (i) rule-based/dictionary/edit-distance; (ii) statistical/frequency/OOV detection; (iii) machine-learning classifiers; (iv) deep-learning sequence or transformer encoders; and (v) hybrid pipelines that combine deterministic filters with learning-based components. The grouping supported cross-study comparison and pattern recognition.

3. RESULTS AND FINDINGS

The synthesis of 35 included studies shows that many pipelines prioritize normalization while under-specifying an explicit detection stage. Where reported, identification methods cluster into five technique types and target a recurrent set of noise categories. Coverage remains English-centric, with fewer resources for low-resource/code-mixed contexts; this affects generalizability and comparability across studies.

3.1. Overview of Included Studies

Most studies addressed social-media or short informal text. A minority described a transparent, standalone identification layer preceding normalization; the remainder embedded detection within correction modules. Reported evaluation practices varied, complicating cross-study benchmarking.

3.2. Technique Categories Observed

Across the corpus, five Categories of identification methods recur. Rule-based, dictionary, and edit-distance approaches are commonly applied as first-pass filters, proving efficient for obvious misspellings, repeated-character elongations, and standard abbreviations. Statistical, frequency, and out-of-vocabulary (OOV) methods flag atypical tokens by leveraging frequency profiles, lexicon coverage, and occasionally phonetic similarity, though they can be sensitive to domain drift. Machine-learning classifiers (at token- or span-level) distinguish clean from noisy tokens and, where annotated data exist, assign specific noise labels. Deep-learning techniques, including sequence models and transformer-based encoders, capture contextual cues and are particularly effective for complex phenomena such as slang and code-mixing. Finally, hybrid pipelines integrate deterministic filters with ML/DL components to balance precision, recall, and interpretability, often yielding the most robust performance across heterogeneous social-media text [6].

3.3. Consolidated Noisy-Text Categories

The most frequently addressed categories are (i) misspellings/typographical errors; (ii) abbreviations/contractions; (iii) phonetic or creative spellings; (iv) character elongation; (v) slang and informal expressions; (vi) code-mixing and transliteration; and (vii) domain-specific OOV tokens [7]. Studies that explicitly label categories report clearer routing of tokens to appropriate handlers and more interpretable results.

3.4. Language Coverage and Low-Resource Contexts

Evidence is predominantly English-centric. Limited work covers Malay or other low-resource languages, where short forms, creative spellings, and code-mixing are common. Explicit identification steps are particularly valuable in these contexts to prevent over- or mis-correction and to maintain semantic fidelity.

3.5. Cross-Study Challenges

Three challenges recur: (i) scarcity of fine-grained annotated corpora with explicit noise-type labels; (ii) inconsistent taxonomies and definitions of noise categories; and (iii) limited multilingual benchmarks,

which restrict fair comparison and transferability. Reporting standards for the identification stage remain uneven.

4. CONCLUSION

The review demonstrates that explicit noisy-text identification is crucial yet under-represented in current NLP literature. Future research should formalize the identification stage, establish standard taxonomies, and create open Malay-English datasets. Integrating rule-based precision with deep-learning robustness can enhance normalization pipelines and cross-lingual transferability [8].

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LEVERAGING ARTIFICIAL INTELLIGENCE FOR ACCELERATED VACCINE DISCOVERY: A DATA-DRIVEN ANALYSIS OF A CRITICAL HOSPITAL SUPERBUG

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ABSTRACT -The escalating threat of multidrug-resistant (MDR) hospital superbugs necessitates a paradigm shift from traditional, time-consuming vaccine development to more rapid, precise and data-intensive strategies. This study presents a powerful, AI-driven computational pipeline that accelerates the identification of immunogenic targets against the superbug *Acinetobacter baumannii*. By integrating pan-genomic analysis with advanced machine learning-based immunoinformatic tools, we systematically deconstructed the genomic complexity of this pathogen. Our automatic workflow began with the quality filtering of 6,806 genomes to a curated set of 213, from which pan-genomic analysis identified 1,687 core proteomes. A sophisticated reverse vaccinology protocol, leveraging a cascade of predictive algorithms, prioritized five conserved, essential and immunogenic surface proteins. From these, machine learning models for epitope prediction identified 14 highly antigenic B-cell-derived T-cell epitopes. The final multi-epitope vaccine construct was modelled and validated *in silico* with molecular docking simulations using the HADDOCK 2.4 web server, which confirmed high-affinity binding to Toll-like receptors (TLR-2 and TLR-4). This work underscores the transformative potential of AI-powered bioinformatics in modern vaccinology, offering a scalable and efficient blueprint for combating high-priority pathogens.

Keywords: Artificial Intelligence; Hospital Superbug; Reverse vaccinology; Multi-epitope Vaccine; Data Science

1. INTRODUCTION

Hospital-acquired infections caused by multidrug-resistant (MDR) pathogens, often referred to as “superbugs,” represent a critical and growing threat to global public health. These organisms are notoriously difficult to treat and are associated with increased patient morbidity and mortality [1]. Among the most dangerous of these superbugs is *Acinetobacter baumannii* (*A. baumannii*), a Gram-negative bacterium that has earned a Priority 1 “critical” classification from the World Health Organization due to its extensive resistance to carbapenems and other last-resort antibiotics [2]. With a dwindling pipeline of effective drugs, vaccination stands as a superior long-term strategy; yet no licensed vaccine against *A. baumannii* is currently available, largely due to the pathogen’s genomic diversity and the slow pace of traditional laboratory-based discovery methods [3].

To overcome these hurdles, the field of vaccinology is increasingly turning to data science and artificial intelligence (AI). Computational approaches such as reverse vaccinology (RV) and pan-genomics have revolutionized the discovery process [4]. Instead of culturing pathogens, these *in silico* methods begin with genomic data, utilizing algorithms and machine learning models to mine entire proteomes for proteins with the ideal characteristics of a vaccine candidate: surface exposure, conservation across strains, essentiality for survival and high antigenicity [5].

This study details the application of a comprehensive, AI-driven computational workflow to systematically identify and validate a multi-epitope vaccine candidate against the hospital superbug *A. baumannii*, demonstrating a powerful and replicable strategy for modern vaccine design.

2. MATERIALS AND METHODS

The entire study was conducted *in silico*, employing a multi-phase computational pipeline powered by a suite of specialized bioinformatics and AI tools, as illustrated in Figure 1.

2.1. Phase 1: Genomic Data Curation and Pan-Genome Analysis

The process commenced with the retrieval of 6,816 *A.baumannii* genome assemblies from the National Center for Biotechnology Information (NCBI) database. To ensure high-fidelity input for downstream analysis, this large dataset underwent a rigorous quality control workflow. Genomes were assessed for completion and contamination metrics using CheckM (available at <https://github.com/Ecogenomics/CheckM>) and taxonomic integrity was verified with the Genome Taxonomy Database Toolkit (<https://github.com/Ecogenomics/GTDBTk>). This filtering refined dataset of 213 high-quality, non-redundant genomes. These genomes were then annotated using Bakta pipeline (<https://github.com/oschwengers/bakta>) and the Roary tool (<https://sanger-pathogens.github.io/Roary/>) was employed to construct the pan-genome, identifying a core set of 1,733 genes conserved across at least 99% of the strains.

2.2. Phase 2: AI-Driven Protein Prioritization via Reverse Vaccinology

The core genes were translated into 1,687 proteins, which were fed into a reverse vaccinology pipeline. This phase involved a cascade of filtering steps, each powered by predictive algorithms. First, BLASTp (<https://blast.ncbi.nlm.nih.gov/Blast.cgi>) was used to screen against the human proteome, eliminating proteins with significant homology to minimize the risk of autoimmunity. Next, the subcellular localization of the remaining proteins was predicted using machine learning-based classifier PsortB (<https://www.psort.org/psortb/>) and Gneg-mPLoc (<http://www.csbio.sjtu.edu.cn/bioinf/Gneg-multi/>) to identify surface-exposed candidates. These candidates were further screened for essentiality by querying the Database of Essential Genes (DEG, <http://www.essentialgene.org/>) and for virulence potential against the Virulence Factor Database (VFDB, <http://www.mgc.ac.cn/VFs/>). Finally, the TMHMM server (<https://services.healthtech.dtu.dk/services/TMHMM-2.0/>) predicted transmembrane helices and the ExPASy ProtParam tool (<https://web.expasy.org/protparam/>) calculated physiochemical properties, leading to the prioritization of the five optimal protein candidates.

2.3. Machine Learning- Based Epitope Prediction and Vaccine Modeling

For the five prioritized proteins, a subtractive screening process was initiated to identify the most potent B-cell and T-cell epitopes [6]. The ABCPred server (<http://crdd.osdd.net/raghava/abcpred/>), which utilized a recurrent neural network, predicted potential B-cell epitopes. These were then analyzed for their ability to act as T-cell epitopes using the Propred (MHC-II, <https://webs.iitd.edu.in/raghava/propred/>) and Propred-I (MHC-I, <https://webs.iitd.edu.in/raghava/propred1/>) servers. A suite of AI-powered tools was used for final validation [7]: the Vaxijen server (<http://www.ddg-pharmfac.net/vaxijen/VaxiJen/VaxiJen.html>) scored antigenicity, VirulentPred (<https://bioinfo.icgeb.res.in/virulent2/>) assessed virulence potential, and the IFNepitope server (<http://crdd.osdd.net/raghava/ifnepitope/>) predicted the capacity to induce a gamma-interferon response. This multi-parameter screening yielded 14 final epitopes (MHC I and II). These epitopes were linked together with appropriate adjuvants and linkers to form a multi-epitopes vaccine construct. The three-dimensional structure of this construct was then molded using I-TASSER server (<https://zhanggroup.org/I-TASSER/>) and refined with GALAXYRefine (<http://galaxy.seoklab.org/>). The final validation step involved molecular docking with the HADDOCK 2.4 web server (<https://wenmr.science.uu.nl/haddock2.4/>) to simulate the interaction between the vaccine construct and host immune receptors.

3. RESULTS AND FINDINGS

The application of our AI-driven pipeline successfully and rapidly distilled a massive amount of genomic data into a highly refined and promising vaccine candidate. The workflow from initial data retrieval to the final list of epitopes is visually summarized in Figure 1. The initial dataset of 6,816 genomes was systematically reduced to 1,687 core proteins, which formed the search space for vaccine candidates. The stringent, multi-layered reverse vaccinology protocol efficiently narrowed this down to

five surface proteins with ideal vaccine candidacy profiles: peptidoglycan DD-metalloendopeptidase, M23 family metallopeptidase, an efflux transporter outer membrane subunit, a TonB-dependent receptor, and a TolC family outer membrane protein. These proteins are integral to bacterial survival and pathogenesis, making them excellent targets [8].

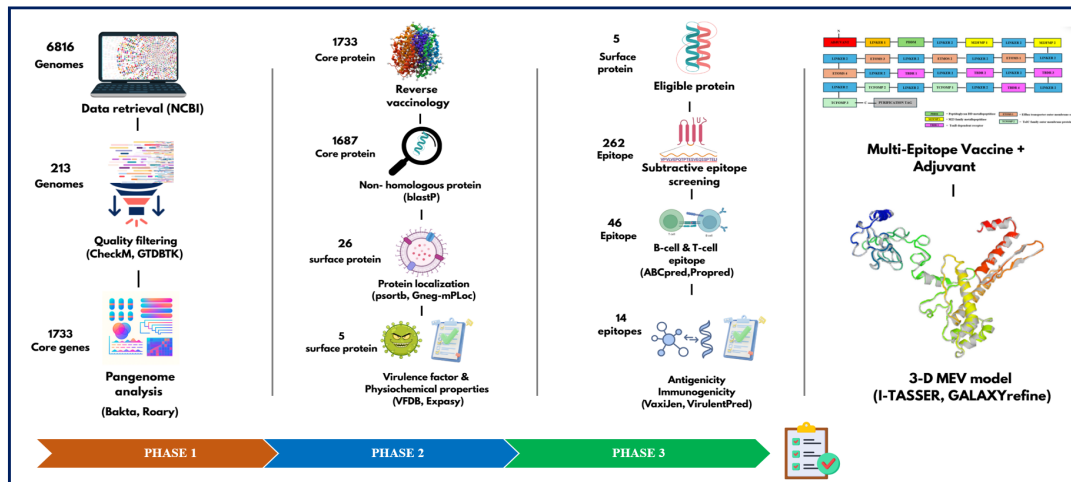


Figure 1. A schematic of the multi-phase computational pipeline. Phase 1 involved genomic retrieval and pan-genomic analysis. Phase 2 utilized a reverse vaccinology cascade to prioritize protein candidates. Phase 3 employed machine learning for epitope screening and 3D modelling of the final multi-epitope vaccine construct.

The final stage of the pipeline, subtractive epitope screening, identified 14 epitopes with high predicted antigenicity, immunogenicity and virulence, while being non-homologous to human proteins and non-allergenic. The computationally designed multi-epitope vaccine construct was then subjected to molecular docking simulation using HADDOCK 2.4 server to predict its interaction with key host immune receptors. As shown in Figure 2, the vaccine construct demonstrated high-affinity binding with both Toll-like receptor 4 (TLR4) and Toll-like Receptor 2 (TLR2). The detailed interaction diagrams revealed multiple hydrogen bonds and non-covalent interactions at the binding interfaces indicating a stable and robust complex. This strong binding is a critical prerequisite for initiating the innate immune signalling cascade that leads to a potent adaptive immune response [9, 10].

4. CONCLUSION

This study powerfully illustrated the capacity of an integrated, AI-driven computational framework to navigate the complexities of pathogen genomics and accelerate modern vaccine design. By systematically applying a series of specialized bioinformatics and machine learning tools, we successfully progressed from thousands of whole genomes to a validated, multi-epitope vaccine candidate for the critical hospital superbug *A.baumannii*. The pipeline not only identified highly conserved and virulence proteins but also pinpointed specific epitopes capable of eliciting a multi-faceted immune response. The final in silico validation through molecular docking provides strong evidence for the construct's ability to effectively engage the host immune system. This data-driven approach serves as a robust, efficient and scalable blueprint for the rapid development of a vaccine against emerging infectious diseases, paving the way for future experimental validation [11].

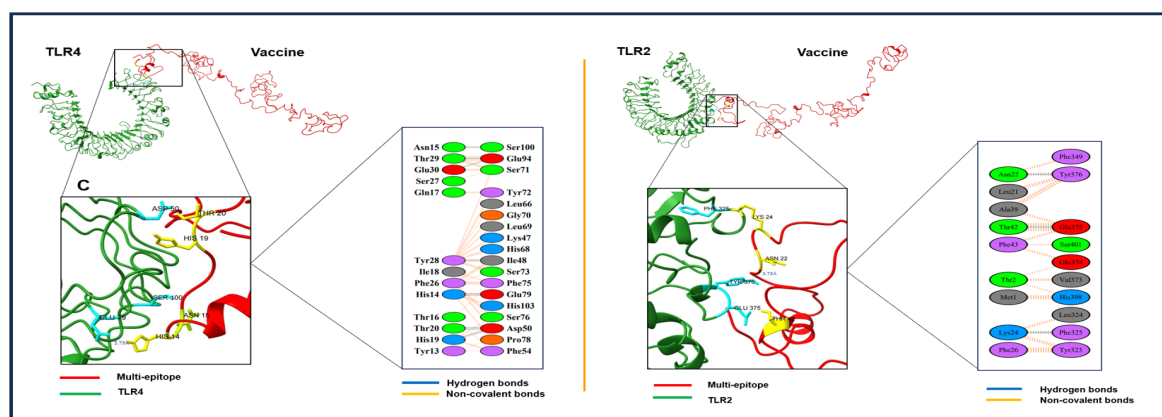


Figure 2. Molecular docking simulation results from the HADDOCK 2.4 server. The multi-epitope vaccine construct (red) is shown binding to TLR4 (left) and TLR2 (right). Zoomed-in panels illustrate the specific amino acid residues involved in the interaction, highlighting the formation of stable hydrogen bonds and non-covalent bonds.

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CNN-BASED CHRONIC DERMATITIS DETECTION AND NON-STEROIDAL PRODUCT RECOMMENDATION

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Abstract. Chronic dermatitis, encompassing conditions such as eczema and psoriasis, affects millions globally and imposes a significant healthcare and economic burden. Traditional diagnosis methods rely heavily on visual examination and clinical expertise, which can lead to misdiagnosis or treatment delays. This study presents a Convolutional Neural Network (CNN)-based system designed to automatically identify chronic dermatitis from digital skin images and recommend safe, non-steroidal treatment alternatives. By leveraging transfer learning and deep neural architectures such as EfficientNet and MobileNet, the proposed system improves diagnostic precision while promoting patient safety through evidence-based product recommendations. Findings from recent studies show CNNs achieving diagnostic accuracies above 95%, indicating the feasibility of integrating AI into teledermatology applications. This framework aims to support early detection, reduce corticosteroid dependence, and improve accessibility to dermatological care, especially in underserved regions.

Keywords: Convolutional Neural Network (CNN); Chronic Dermatitis; Eczema; Psoriasis; Deep Learning; Non-Steroidal Treatment; Teledermatology

1. INTRODUCTION

Chronic dermatitis, which includes diseases like eczema and psoriasis, represents a major dermatological problem worldwide, affecting millions of people. Although much progress is being achieved in medicine, diagnosis is still difficult to conduct well and on time, based merely on subjective visual inspection. In many parts of the world, shortages of dermatologists further exacerbate these problems, which can frequently lead to self-diagnosis by patients or to the misuse of corticosteroids. Although corticosteroids are effective in controlling inflammation, their long-term use may cause side effects such as skin thinning and dependency. For this reason, the need for non-invasive, objective, and accessible diagnostic devices has never been so relevant.

AI, especially CNNs, has shown outstanding performance in the analysis of medical images. This approach allows CNNs to identify complex spatial characteristics from dermatological images on their own and thus increase the consistency of diagnoses. References such as Hassan et al. (2022) and Kim & Park (2023) indicate that the accuracy of CNN-based skin disease classification exceeds 95%. Moreover, transfer learning has enabled pre-trained models such as EfficientNet and MobileNet to fine-tune on specific diagnostic tasks at a reduced computational cost. The integration of such technology into healthcare may enhance accessibility and clinical decision-making with respect to diagnosis.

2. METHODOLOGY

This study adopts a deep learning-based approach using CNN architecture for the classification of chronic dermatitis. The methodology is divided into four major stages: image preprocessing, model training, classification, and recommendation.

In the preprocessing step, images from public datasets like DermNet are enhanced by normalization, color correction, and segmentation to remove noise. Different CNN architectures, such as EfficientNet-B7, MobileNet, and ResNet50, are fine-tuned by transfer learning for classification. Softmax activation

is used for multi-class output for eczema, psoriasis, and normal skin. The metrics used to evaluate the performance will include accuracy, precision, and recall.

The last module covers the development of evidence-based, non-steroidal product recommendations, such as emollients, barrier-repair creams, and natural anti-inflammatory formulations, based on dermatological literature. Recommendations should be condition-specific to enhance safety and treatment personalization.

3. KEY FINDINGS FROM LITERATURE

Various recent studies have shown the potential of CNNs in revolutionizing dermatological diagnosis by automatic image analysis. Early works such as Hassan et al., 2022, showed that an 11-layer CNN could classify dermatological conditions like eczema, hives, and psoriasis with an accuracy of 96%. They had, in essence, managed to overcome the subjectivity of traditional visual examinations using deep feature extraction. Along these same lines, Kim & Park (2023) discussed the use of transfer learning for CNN architectures such as MobileNet and Xception, noting efficiency gains without accuracy deterioration. Their fine-tuning of pre-trained models on medical image datasets has demonstrated how computational demands are significantly reduced-a great advantage with respect to clinical or mobile applications.

Li and Tan (2022) further used CNN to identify eczema areas with impressive accuracy of up to 98.6% by analyzing textures and segmenting them. The work thus points to the benefits of integrating conventional image processing techniques, such as Gray-Level Co-occurrence Matrix (GLCM), into CNN learning to enhance the generalization ability of models. Similarly, Malik et al. (2023) showed that CNN architectures, including EfficientNet-B7, are capable of performing preliminary image analysis even when taken using a mobile device.

Recent works from 2023 to 2025 indicate a trend toward incorporating emerging technologies such as transformers and federated learning into CNNs. Ahmad et al. (2024) proposed a hybrid CNN-transformer network that outperformed previous state-of-the-art approaches by 4% with increased contextual awareness in lesion segmentation. Similarly, Chua and Nguyen (2023) researched lightweight CNNs for on-device mobile dermatology diagnostics, focusing on model performance optimization without sacrificing inference speed. Gao et al. (2025) then went on to propose the use of federated CNN frameworks where model training can be done with decentralized data privacy, having multiple healthcare institutions advance in model training without compromising data privacy-a very crucial development for medical ethics and patient confidentiality.

Despite these developments, some issues still have not been addressed in the literature. Most of the CNN models focus on classification accuracy without considering post-diagnostic support, such as recommending a suitable treatment or counseling patients. Furthermore, dataset bias and poor representation concerning skin tones take their toll on model fairness and reliability. These shortcomings pinpoint the research deficit that this project falls under: developing a CNN-based model that will identify chronic dermatitis conditions and include intelligent, evidence-based non-steroidal treatment recommendations personalized for each patient.

4. PROPOSED FRAMEWORK

The proposed architecture for CNN-Based Chronic Dermatitis Detection and Non-Steroidal Product Recommendation will involve three fundamental building blocks: image preprocessing, CNN-based classification, and intelligent recommendation. Each of these components acts as an integrated layer within one pipeline aimed at accuracy, interpretability, and practical feasibility.

During the preprocessing of the images, raw input images are transformed to a standard quality and artifact-free images. This includes resizing, normalizing, color correction, and segmentation. Segmentation is especially important because it separates the area of the lesion from healthy skin around

it, ensuring that the CNN focuses on features that are relevant. U-Net and MAGRes-UNet architectures can be employed in this stage, inspired by Arora et al. 2023 and Javed et al. 2023, respectively, to maintain boundary definitions and extract multi-scale contextual information. Preprocessing aids further in reducing the effects of uneven illumination and variable resolutions of images besides skin tone differences, factors which are known to reduce diagnostic accuracy when datasets are unstandardized.

The CNN-based classification module acts as the analytic core of the system. It proposes to fine-tune well-known, pre-trained CNN architectures like EfficientNet-B7, MobileNetV3, and ResNet50 on curated eczema, psoriasis, and healthy skin images via transfer learning. Each model layer contributes to the complex process of feature extraction, where differences in texture, color, and patterns that characterize each condition are learned step by step. The Softmax activation function at the last layer gives class probabilities, enabling interpretability along with diagnostic confidence levels. Besides, explainable AI tools such as Grad-CAM visualization may be used to highlight the most important regions of the image contributing to the model's prediction, enhancing trust and clinical interpretability.

The Recommendation Module constitutes the novel contribution of this research. Instead of furnishing a mere diagnosis, the system assembles a list of clinically validated, non-steroidal treatment options tailored to the detected condition. These might include emollients, ceramide-based moisturizers, calcineurin inhibitors, or plant-derived anti-inflammatory products supported by current dermatological research. The recommendations will be adaptive, factoring in user inputs like skin sensitivity, condition severity, or response to previous treatments. This personalized output ensures much safer management strategies, minimizing overdependence on corticosteroids.

The system proposed is designed with scalability and integration in mind. The prototype can be either web-based or a mobile application, which enables the user to capture an image through smartphone cameras for immediate analysis. This provides not only an improvement in the availability of diagnostics but also fits into the more general trend in health care toward preventive and patient-centered care. Furthermore, by its modular design, future integrations with either cloud computing or federated learning frameworks are possible, ensuring data security with continuous model improvements due to distributed learning.

5. CONCLUSION

The work fills an important gap in dermatological AI through the introduction of a CNN-based diagnostic framework extending from detection to clinical recommendation. By integrating image analysis with nonsteroidal product guidance, the system offers a dual contribution: enhanced diagnostic precision and the fostering of safer, evidence-based treatment decisions. Including transfer learning, lesion segmentation, and explainable AI enhances the technical strengths of this system while preserving its clinical interpretability. This approach fits within current healthcare priorities-to make high-quality diagnostics available, affordable, and sustainable using AI.

In the context of real-world application, the proposed system holds immense potential for teledermatology and mobile health (mHealth) platforms. A mobile-compatible AI diagnostic assistant can enable patients to obtain early, reliable assessments in resource-limited regions with scarce access to dermatologists. Healthcare professionals will also be able to use the system as a decision-support tool, reducing workload and improving triage efficiency.

The model's modular design further supports future enhancements that will include multimodal inputs-textual symptoms, patient history-and connecting to electronic health records for more holistic patient profiling. Clinical validation and expansion of the dataset are required to ensure the robustness of the model for various populations and skin types. Federated learning will further improve generalizability while maintaining patient data privacy. Product recommendation logic should also be refined in continuous consultation with dermatologists, validating the safety and efficacy of recommendations. This indeed provides the foundation for an intelligent, patient-centric dermatology platform that

incorporates the precision of CNN with the compassion of personalized care, changing how chronic dermatitis is diagnosed and treated around the world.

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ADAPTIVE COGNITIVE SUPPORT FOR ALZHEIMER'S PATIENTS VIA DECISION TREE-BASED MOBILE APPLICATION

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ABSTRACT Existing mobile applications for Alzheimer's care often prioritize general caregiver support while lacking features that actively engage patients in cognitive exercises or track memory performance. Many also suffer from complex interfaces unsuitable for users with cognitive impairments. This project introduces Alzheimer's Ally, a mobile application designed to deliver personalized cognitive support through memory-based quizzes using familiar images and voice prompts. The system employs the Classification and Regression Tree (CART) algorithm to dynamically adjust quiz difficulty based on user performance metrics such as accuracy and response time. A built-in dashboard visualizes performance trends, aiding caregivers in monitoring patient progress. Developed using the Waterfall methodology, the application integrates Flutter, Firebase, and Python to ensure a structured and user-friendly experience. By combining adaptive engagement with personalized content, Alzheimer's Ally aims to enhance cognitive stimulation and improve the quality of life for both patients and caregivers.

Keywords: Alzheimer's; cognitive care; decision tree; mobile application; personalization

1. INTRODUCTION

Cognitive health is essential for maintaining independence and quality of life, particularly among the elderly. Alzheimer's disease, the most common form of dementia, progressively impairs memory, reasoning, and behavior, making daily tasks increasingly difficult. Although there is no cure, studies have shown that regular cognitive stimulation can help delay the decline in mental function [3],[5]. Personalized cognitive care has therefore gained attention as an effective approach to support individuals with Alzheimer's, as it enhances engagement, reduces agitation, and improves overall well-being [5].

Despite the proliferation of mobile health applications, many existing tools focus primarily on caregiver support rather than patient engagement [3]. These applications often suffer from poor user interface design, limited personalization, and inadequate tracking of memory performance. To address these challenges, this study introduces Alzheimer's Ally, a mobile application designed to deliver personalized cognitive support through memory-based quizzes.

The system leverages familiar multimedia content—such as family photos and voice recordings—and uses the Classification and Regression Tree (CART) algorithm to dynamically adjust quiz difficulty based on user performance metrics like accuracy and response time [1],[2],[4]. Additionally, a built-in dashboard provides caregivers with real-time insights into patient progress, enabling them to deliver more informed and responsive care.

2. MATERIALS AND METHODS

The development of Alzheimer's Ally followed the Waterfall software development methodology, chosen for its structured and sequential approach suited to projects with clearly defined requirements and minimal expected changes during development. The application was built using Flutter for the front-end to ensure a responsive, user-friendly interface suitable for elderly users. Firebase was used for backend services, such as authentication, real-time database (Cloud Firestore), and media storage. A Python-based machine learning module was integrated to implement the Classification and Regression Tree (CART) algorithm, which dynamically adjusts quiz difficulty based on user performance metrics such as accuracy and response time [1],[4].

To simulate real-world usage, synthetic data was generated using Python. The dataset included three key attributes: quiz accuracy, average response time, and the predicted difficulty level. Conditional logic was applied to label quiz difficulty as Easy, Medium, or Hard, depending on performance. The CART model was trained using this dataset and achieved a high accuracy of 99.7%, demonstrating its reliability in predicting appropriate quiz levels [2],[4].

The app allows caregivers to upload personalized materials—images and voice recordings of family members—which are used in memory-based quizzes. Each quiz consists of five questions, randomly selected from uploaded content. The CART model evaluates the patient’s previous performance to determine the difficulty of subsequent quizzes. A built-in dashboard visualizes performance trends, enabling caregivers to monitor memory changes over time.

3. RESULTS AND FINDINGS

The CART model implemented in Alzheimer’s Ally was trained using synthetic data that simulated realistic user performance patterns. The dataset included quiz accuracy and average response time, which were used to predict the next quiz difficulty level. After training and evaluation, the model achieved an impressive accuracy of 99.7%, indicating its reliability in adapting quiz difficulty appropriately [1],[4].

The decision tree logic prioritizes assigning a *Medium* difficulty level by default, but adjusts based on performance: *Easy* is assigned when both accuracy is low and response time is slow, while *Hard* is assigned when accuracy is high and response time is fast. This adaptive mechanism ensures that patients are neither overwhelmed nor under-stimulated, promoting consistent engagement [2],[5].

The application features a clean and intuitive interface tailored for elderly users. Key functions include personalized quizzes using uploaded family images and voice prompts, a performance dashboard displaying trends in accuracy and response time, and an album viewer for revisiting familiar faces and voices. A sample dashboard from the app is shown in Figure 1. A total of 21 test cases were executed to validate the system’s functionality, covering user registration, material upload, quiz execution, dashboard analytics, and account management. All test cases passed successfully, confirming the system’s robustness and usability.

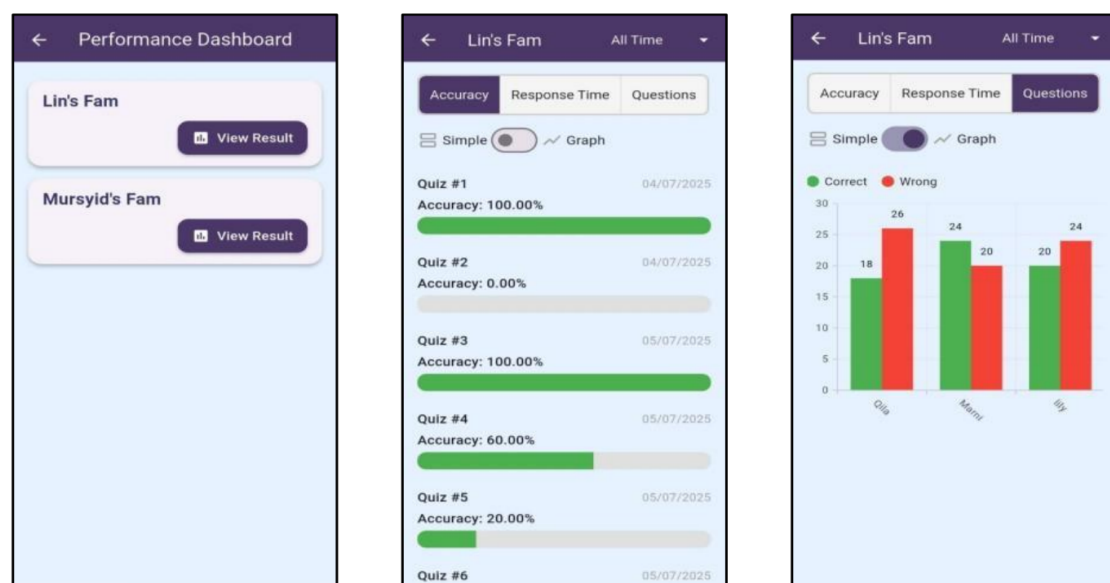


Figure 2. Sample dashboard view

4. CONCLUSION

This project introduced Alzheimer's Ally, a mobile application designed to provide personalized cognitive support for individuals with Alzheimer's disease. By integrating familiar multimedia content and adaptive quiz mechanisms powered by the CART decision tree algorithm, the application offers a meaningful and engaging experience tailored to each patient's cognitive ability. The system demonstrated high predictive accuracy (99.7%) in adjusting quiz difficulty based on user performance, ensuring that patients remain mentally stimulated without being overwhelmed. The inclusion of a performance dashboard allows caregivers to monitor progress and adapt care strategies accordingly. Functionality testing confirmed the reliability and usability of the application, with all core features operating as intended. While the current version is limited to Android devices and requires internet connectivity, future enhancements could include multi-language support, offline capabilities, and broader platform compatibility. Overall, Alzheimer's Ally represents a promising step toward technology-driven, personalized cognitive care, with the potential to improve quality of life for both patients and caregivers.

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ADVANCES IN IMAGE CAPTIONING FOR IMAGE RETRIEVAL: AN OVERVIEW OF GENERATION AND RETRIEVAL-AUGMENTED MODELS

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ABSTRACT - Image captioning has emerged as an essential component in improving image retrieval by bridging the gap between visual content and natural language. Traditional retrieval techniques depend heavily on low-level visual features or manually annotated tags, which often fail to capture semantic meaning. By integrating image captioning, retrieval systems can interpret visual content more effectively and enhance the accuracy and flexibility of semantic search across complex datasets such as medical imaging, remote sensing, and social media archives. Recent advancements in deep learning, particularly encoder-decoder architectures, attention mechanisms, and transformer-based models, have significantly improved caption fluency and contextual understanding. This overview highlights two main paradigms of image captioning for retrieval; generation-based and retrieval-augmented approaches, outlining their mechanisms, strengths, and limitations. Retrieval-augmented models reuse captions from similar images to improve efficiency and grammatical accuracy, while generation-based methods employ CNNs, RNNs, or transformers to generate descriptive and semantically rich captions. Although these approaches have improved retrieval accuracy, challenges such as semantic misalignment, semantic inconsistency, and domain adaptation remain. Addressing these issues requires continued exploration of adaptive, hybrid, and retrieval-augmented models to enhance semantic alignment, robustness, and generalization across diverse retrieval scenarios.

Keywords: Image captioning, Image retrieval, Deep learning, Generative models, Retrieval-augmented model

1. INTRODUCTION

Image captioning is the process of generating natural language descriptions for images, has become a critical enabler for advanced image retrieval systems. Traditional retrieval techniques rely primarily on low-level visual features or manually annotated tags, which often fail to capture semantic meaning [1]. By integrating image captioning, retrieval systems can interpret visual content semantically, thereby improving retrieval relevance and flexibility across complex datasets such as remote sensing, medical imaging, and social media archives. Image captioning enables the systems to generate descriptive text for images and use these captions to improve search accuracy and semantic understanding [2]. Recent research integrates deep learning, and external memory to bridge the gap between visual content and natural language, enhancing both the quality of captions and the effectiveness of retrieval

Recent advances in deep learning have led to significant progress in image captioning, particularly with encoder-decoder architectures, attention mechanisms, and transformer-based models [3][4]. These approaches have not only improved caption fluency but also bridged the semantic gap between vision and language. Besides, retrieval-augmented and hybrid models have been proposed to integrate external memory or previously retrieved captions as contextual guidance, further enhancing caption quality and retrieval precision [5][6]. This paper discusses the present approaches and applications of image captioning for retrieval, highlighting key methods, and their effectiveness in enhancing semantic search and retrieval accuracy.

2. APPROACHES TO IMAGE CAPTIONING FOR RETRIEVAL

2.1. Generation-Based Methods

Generation-based methods employ encoder–decoder frameworks, typically using convolutional neural networks (CNNs) for visual encoding and recurrent neural networks (RNNs) or transformers for caption generation, especially for tasks involving image, text, and signal generation or transformation [7][8][9]. These approaches have become a standard due to their strong ability to extract and reconstruct complex features. Encoder-decoder frameworks (often using CNNs and RNNs or Transformers) generate new captions tailored to the input image, capturing complex semantics and relationships. Figure 1 shows an image segmentation network architecture based on encoder-decoder from [10]. The diagram illustrates an image processing pipeline where input images are encoded to extract features, decoded to reconstruct meaningful representations, and finally produce prediction outputs for image captioning.

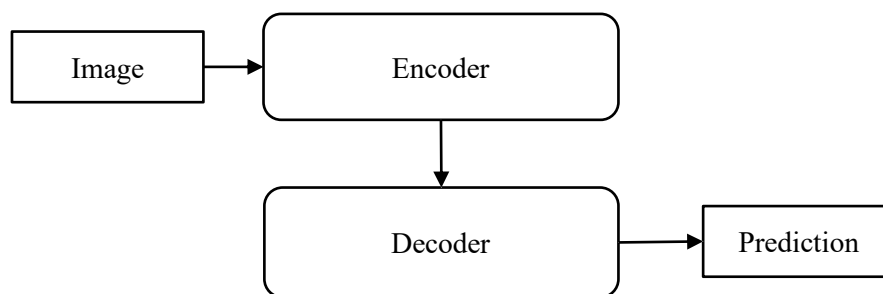


Figure 1. An image segmentation network architecture based on encoder-decoder structure [10]

The encoder, typically a CNN, extracts high-level features from input data (such as images or signals), compressing them into a latent representation. The decoder, often a recurrent neural network (RNN) like LSTM or another CNN, reconstructs or generates the target output from this representation. CNN-based encoder–decoder models excel at capturing spatial hierarchies, enabling pixel-level predictions and dense output generation. They are also adaptable, supporting integration with attention mechanisms, transformers, and Bayesian approaches for uncertainty quantification [10][7]. However, [9] stated that, these models may generate descriptions of objects that do not exist in the image and require large-scale annotated dataset for effective training, therefore their performance often depends heavily on dataset quality and diversity, and they may struggle to generalize effectively to unseen domains or real-world scenarios.

2.2. Retrieval-Based Methods

Retrieval-based methods for image captioning have gained significant attention as they leverage existing annotated data to enhance the quality and relevance of generated captions. These approaches often combine retrieval of similar images or captions with generative models, aiming to bridge the gap between visual content and natural language descriptions, where these methods rely on matching a new image with similar images in a database and reusing their captions. This approach typically involves two main steps: (1) retrieving visually or semantically similar images and their captions from a large dataset, and (2) using these retrieved captions as direct outputs or as auxiliary information to guide generative models. Modern methods often integrate retrieval with deep learning architectures, such as transformers or memory-augmented networks, to improve contextual understanding and caption diversity [5][6][11].

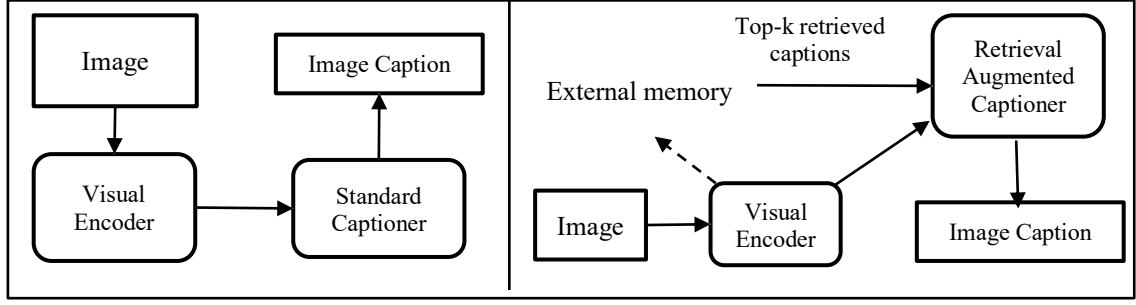


Figure 2. Comparison between a standard captioner (left) and the retrieval-augmented captioning schema (right), in which an external memory is employed to condition the generation process. [6]

Figure 2 above illustrates the comparison between a standard captioner and the retrieval-augmented captioning approach. The left shows a standard captioner generating captions directly from image features. The right adds external memory to retrieve top-k similar captions, improving caption generation with additional contextual knowledge. This shows that retrieval-augmented captioning systems efficiently generate fluent and grammatically correct captions using knowledge from external memory. However, they struggle with unseen images or novel object compositions since they depend heavily on existing captioned image pairs [5]. These systems retrieve captions from a database of existing image-caption pairs based on visual or semantic similarity. While they often produce grammatically correct captions, they may lack specificity or relevance to new or unseen images.

3. DISCUSSION

The discussion demonstrates that retrieval-augmented captioning models and generation-based methods currently represent the most promising and effective strategies for image retrieval. Both approaches leverage deep learning architectures to bridge the semantic gap between visual and textual modalities, thereby enhancing retrieval accuracy and caption diversity. Generation-based methods employ encoder-decoder frameworks, typically using convolutional neural networks (CNNs) for visual encoding and recurrent neural networks (RNNs) or transformers for caption generation. These models produce linguistically coherent and descriptive sentences that capture the semantic content of an image, enabling more meaningful and interpretable retrieval results. Retrieval-augmented captioning models, on the other hand, combine generative capabilities with external knowledge or retrieved captions, allowing the system to generate more contextually grounded and semantically enriched descriptions. This integration of retrieval and generation helps to overcome limitations of purely generative systems by enhancing factual accuracy and robustness across visual domains.

Despite their effectiveness, several challenges remain in achieving consistent and reliable performance in real-world applications. One major issue is semantic misalignment, where retrieved captions may contain irrelevant or biased information, leading to inaccurate or misleading description as discussed in [11]. Semantic inconsistency and data dependence are also persistent problems, as generative models often describe inaccurate or visually inconsistent details or rely excessively on large annotated datasets for effective training [9]. This dependency limits their scalability and adaptability to unseen or low-resource domains. Addressing these limitations requires more adaptive architectures, diversified datasets, and improved cross-domain learning mechanisms to ensure that captioning models remain semantically accurate, context-aware, and generalizable across different retrieval scenarios.

In summary, each approach offers distinct strengths and limitations. Generation-based models excel in producing flexible and semantically rich captions but often require large-scale annotated data and may generate descriptions of objects that do not exist in the image. Retrieval-augmented methods enhance factual accuracy and contextual grounding through external memory, yet they depend heavily on existing caption databases and struggle with unseen visual concepts. Understanding these trade-offs is essential for developing more adaptive and generalizable image captioning systems.

4. CONCLUSION

This study highlights the vital role of image captioning in advancing image retrieval by enabling systems to understand and describe visual content through natural language. The integration of captioning enhances retrieval relevance and semantic understanding, surpassing traditional methods that rely solely on low-level visual features or manual annotations. The study identifies two major approaches, the retrieval-based and the generation-based methods as the foundation for modern captioning systems. Retrieval-based approaches demonstrate efficiency and grammatical accuracy by reusing existing captions, while generation-based methods, using encoder–decoder frameworks with CNNs, RNNs, or transformers, provide flexibility and produce contextually rich captions. The combination of these techniques has led to improved caption quality, diversity, and retrieval precision across diverse datasets and domains. However, significant challenges persist, particularly in maintaining semantic alignment, reducing semantic inconsistency, and achieving better domain adaptation. Retrieved captions may introduce irrelevant or biased information, while generative models often depend on large annotated datasets and may produce inaccurate or irrelevant descriptions. Therefore, future captioning research should further investigate and evaluate hybrid and retrieval-augmented models, emphasizing improvements in semantic accuracy, contextual robustness, and adaptability for real-world image retrieval applications.

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UNVEILING KNOWLEDGE CONCEALMENT IN MALAYSIA'S PUBLIC SECTOR: THE ROLES OF REWARDS, INTERNAL COMPETITION, AND INTERPERSONAL DISTRUST

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ABSTRACT Knowledge sharing is crucial for the public sector's efficiency and innovative capabilities, but is often hindered by practices of knowledge concealment (KC). This study sets out to examine how KC is driven by the lack of rewards in sharing knowledge and internal competition and mediated by interpersonal distrust. Grounding on the Social Exchange Theory (SET), online surveys were distributed to 276 *Pegawai Tadbir dan Diplomati* (PTD), located in the Klang Valley. Upon analyzing the data using Partial Least Squares Structural Equation Modelling (PLS-SEM), it was found that KC is positively and significantly driven by the lack of rewards in knowledge sharing and internal competition. The mediating result showed the positive and significant role of interpersonal distrust in the relationship between KC and the lack of rewards in knowledge sharing, but otherwise in the relationship between KC and internal competition. Theoretically, these results expand the SET by elucidating the negative effect of distrust and perceived inequity on knowledge sharing. The results suggest that knowledge concealment behaviors can be mitigated by establishing fair reward mechanisms and management based on trust. This study promotes knowledge-based and innovative public governance in the country, supporting SDGs 8 and 9 on Decent Work & Economic Growth and Industry, Innovation, & Infrastructure.

Keywords: Knowledge concealment; Interpersonal distrust; Lack of rewards in knowledge sharing; Internal competition; Malaysia.

1. INTRODUCTION

Knowledge sharing is a crucial resource for improving the innovation, efficiency, and competitiveness of organizations. But the full impact of this resource is often hampered by knowledge concealment (KC) [1,2]. Previous studies have proven that KC is a problem for both the private and public sectors, but even more so in the latter, as it is highly hierarchical, poorly incentivized, and vastly bureaucratized, hindering open communication and collaborative learning [3-5]. Consequently, these organizations suffer tremendously from this practice as it weakens collaboration, trust, and innovative capabilities [6].

While prior studies have linked KC to leadership style and motivation, there remains limited empirical evidence on how it is influenced by organizational and relational factors in highly bureaucratic settings [8]. In Malaysia, *Pegawai Tadbir dan Diplomati* (PTD) are public service officers who play a key role in Malaysia's policy development and implementation [7]. Yet, their knowledge sharing and service delivery efforts are often constrained by issues such as a lack of rewards in knowledge sharing, internal competition, and interpersonal distrust.

Despite the growing attention to KC, few studies have examined the mediating role of interpersonal distrust within this context. Hence, this study aims to examine how KC among PTD officers is driven by the lack of rewards in knowledge sharing and internal competition and mediated by interpersonal distrust. By elucidating the social functions of perceived inequity and distrust in driving knowledge concealment behaviors, this study is expected to expand the bounds of the Social Exchange Theory (SET).

2. MATERIALS AND METHODS

A quantitative and cross-sectional research design was employed to determine the direct and mediated relationships in this study, involving the constructs of lack of rewards, internal competition,

interpersonal distrust, and knowledge concealment. A structured questionnaire was disseminated online to a group of *Pegawai Tadbir dan Diplomati* (PTD), in which 276 responses were received. To ensure that all the relevant ministries and departments were well represented, stratified purposive sampling was utilized. All the measurement items were adapted from past works, specifically [1] for knowledge concealment, [10] for interpersonal distrust, and [11] for lack of rewards and internal competition. A 5-point Likert scale was used to rate the items, with 1 = strongly disagree and 5 = strongly agree. Partial Least Squares Structural Equation Modelling (PLS-SEM) with SmartPLS 4.0 was employed for the data analysis, testing the direct and mediated effects according to the set procedures [12].

3. RESULTS AND DISCUSSION

Based on the Partial Least Squares Structural Equation Modelling (PLS-SEM) analysis results, knowledge concealment is strongly and significantly predicted by lack of rewards ($\beta = 0.555$, $p < 0.001$), and is moderately but significantly affected by internal competition ($\beta = 0.127$, $p < 0.05$). It is also positively and significantly affected by interpersonal distrust ($\beta = 0.243$, $p < 0.001$). Meanwhile, interpersonal distrust is positively and significantly linked to lack of rewards ($\beta = 0.314$, $p < 0.001$), but is insignificantly linked to internal competition ($\beta = -0.036$, $p = 0.345$).

In terms of the mediating effect, interpersonal distrust is found to mediate the pathway between lack of rewards and knowledge concealment ($\beta = 0.076$, $p < 0.001$). However, it has no mediating effect in the link between internal competition and knowledge concealment ($\beta = -0.009$, $p = 0.344$). These results indicate that without incentives or recognition for sharing knowledge, employees tend to develop distrust and hence will withhold any needed information. On the contrary, internal competition at moderate levels does not drive distrust in this context.

Consistent with the assertion of the Social Exchange Theory [9], these findings show that employees reciprocate what they receive from their employer. When sharing knowledge is perceived to be a thankless or one-sided practice, employees would rather keep that knowledge to themselves [13, 14]. This study confirms past insights regarding the key role of distrust in driving the impact of organizational factors on knowledge concealment, especially in the public sector.

Table 1. Structural Model Results for Direct and Indirect Effects

Path	Std. Beta	p-value	Significant	Effect type
Lack of Rewards in Knowledge Sharing → Knowledge Concealment	0.555	<0.001	***	Direct
Internal Competition → Knowledge Concealment	0.127	0.043	*	Direct
Interpersonal Distrust → Knowledge Concealment	0.243	<0.001	***	Direct
Lack of Rewards in Knowledge Sharing → Interpersonal Distrust	0.314	<0.001	***	Direct
Internal Competition → Interpersonal Distrust	-0.036	0.345	ns	Direct
Lack of Rewards in Knowledge Sharing → Interpersonal Distrust → Knowledge Concealment	0.076	<0.001	***	Indirect
Internal Competition → Interpersonal Distrust → Knowledge Concealment	-0.009	0.344	ns	Indirect

4. CONCLUSION

It is concluded that knowledge concealment among PTDs is driven by the lack of rewards and interpersonal distrust, representing organizational and relational factors in general. Theoretically, this study expands the SET by showing how lack of rewards for sharing knowledge can be perceived as a practice of inequity, generating distrust. As a suggestion, to better understand concealment behaviors in the growing digital landscape, future studies could examine cross-cultural distinctions, long-term behavioral changes, and trust-building mechanisms based on AI.

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BIBLIOMETRIC ANALYSIS OF AUTOMATED ESSAY SCORING (AES) IN HIGHER EDUCATION (2010–2025) BASED ON SCOPUS DATA

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ABSTRACT Automated Essay Scoring (AES) uses artificial intelligence (AI) and Natural Language Processing (NLP) to automatically evaluate and score essays. In higher education, it helps make grading faster and more consistent. AES systems analyze grammar, vocabulary, and content relevance using techniques from rule-based models to neural networks. However, issues such as fairness, bias, and reliability remain important challenges. This study presents a bibliometric analysis of AES research in higher education from 2010 to 2025 using Scopus data. A total of 224 publications were analyzed with VOSviewer to identify major trends, active authors, and key themes. The findings show steady growth in AES research, especially with the rise of large language models (LLMs) and AI-driven feedback tools.

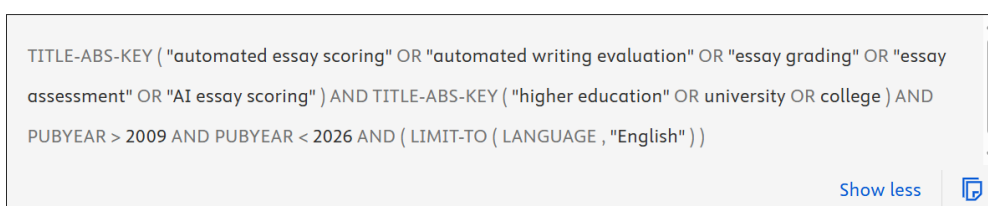
Keywords: Automated Essay Scoring (AES); Artificial Intelligence; Natural Language Processing; Higher Education; Bibliometric Analysis; VOSviewer; Research Trends.

1. INTRODUCTION

Automated Essay Scoring (AES) uses computer algorithms and artificial intelligence (AI) to evaluate and score written essays. It works by using Natural Language Processing (NLP) to check things like grammar, meaning, and how well the essay answers the question. AES systems apply different techniques, from simple machine learning models to more advanced neural networks, to make the scoring process more accurate and similar to human grading. [1][2][3] In higher education, AES is especially helpful because it can handle a large number of essays quickly and consistently, saving time and reducing the workload for educators. [4][5] However, AES still faces challenges such as ensuring fairness, reliability, and understanding creativity or context as humans do [6][7]. Bias in training data can also affect scoring, which makes fairness an important issue to improve. [8][9] To make AES more meaningful, researchers are now working on systems that can give detailed feedback, not just scores, to help students improve their writing. [8][9] Bibliometric analysis is a useful method to study research trends and developments in this area. By using tools like VOSviewer to analyze publications and keywords, it helps identify key studies, popular topics, and research gaps. [10][11][12] Overall, AES has grown into an important tool in education, and bibliometric analysis helps us understand how this field has developed and where it can go next.

2. MATERIALS AND METHODS

Bibliographic data were collected from the Scopus database on 23 October 2025. The search targeted article titles, abstracts, and keywords using the following query:



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TITLE-ABS-KEY ( "automated essay scoring" OR "automated writing evaluation" OR "essay grading" OR "essay assessment" OR "AI essay scoring" ) AND TITLE-ABS-KEY ( "higher education" OR university OR college ) AND PUBYEAR > 2009 AND PUBYEAR < 2026 AND ( LIMIT-TO ( LANGUAGE , "English" ) )
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Figure 1 shows a screenshot of a Scopus search query box. The query is: TITLE-ABS-KEY ("automated essay scoring" OR "automated writing evaluation" OR "essay grading" OR "essay assessment" OR "AI essay scoring") AND TITLE-ABS-KEY ("higher education" OR university OR college) AND PUBYEAR > 2009 AND PUBYEAR < 2026 AND (LIMIT-TO (LANGUAGE , "English")). There is a "Show less" button and a Scopus logo at the bottom right of the box.

Figure 1. Query used to generate data from Scopus database

The search was limited to English-language publications from 2010 to 2025, including journal articles, conference papers, and review papers. The data were then organized and cleaned using Microsoft Excel to remove duplicates and prepare the dataset for analysis. After that, VOSviewer software was used to visualize and analyze the data, including keyword co-occurrence, author collaboration, and citation networks. Keywords that appeared at least five times were included to identify the main research themes.

3. RESULTS AND FINDINGS

The results of this bibliometric analysis reveal significant trends and patterns in the development of Automated Essay Scoring (AES) research in higher education. Key findings encompass publication growth, major contributors, and thematic clusters that illustrate the field's progression from early rule-based systems to AI-driven approaches.

3.1. Publication Trends

A total of 224 publications were found in this period, showing steady growth in AES research.

- 2010–2015 (Early Stage): Research was still limited and exploratory. Most studies focused on basic models and testing the potential of AES systems.
- 2016–2020 (Growth Phase): The number of studies started to increase, especially after 2018, when neural networks and deep learning became more common in AES.
- 2021–2023 (Rapid Expansion): Publications and citations increased sharply, influenced by online learning during and after the COVID-19 pandemic and the use of generative AI in education.
- 2024–2025 (Maturity): Research output became stable, showing that AES is now an established topic in AI and education studies.

Table 1. Publication Trends per year

Row Labels	Count of Title	Sum of Cited by	Growth year by year (%)
2010	7	46	100
2011	5	66	-28.57
2012	1	12	-80
2013	10	338	900
2014	8	344	-20
2015	6	342	-25
2016	8	216	33.33
2017	11	267	37.50
2018	6	30	-45.45
2019	13	442	116.67
2020	12	544	-7.69
2021	17	649	41.67
2022	20	318	17.65
2023	23	1403	15
2024	39	284	69.57
2025	38	73	-2.56

3.2. Key Authors and Institutions

The most active authors in AES research include Burstein, J. and Zhang, Z., followed by McCaffrey, D.F., Ge, S., and Link, S. These researchers often collaborate with Educational Testing Service (ETS) in the United States, which is also the most productive and influential institution in the field. ETS has played a big role in the development of AES tools such as the e-rater® system. Other active institutions include the New Jersey Institute of Technology and Duolingo, Inc., showing growing interest from both academic and commercial sectors.

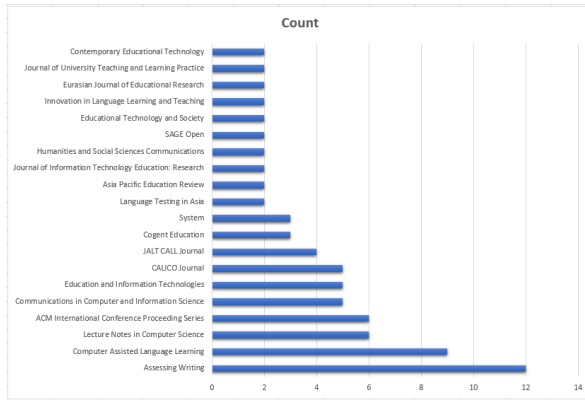


Figure 2. Top Journals

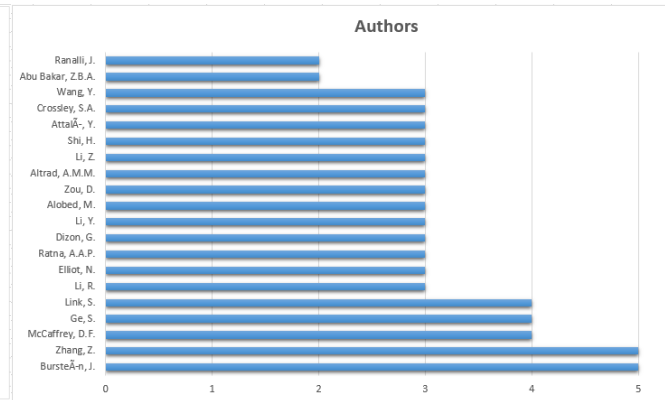


Figure 3. Top Authors

3.3. Countries and Collaboration

The United States leads AES research with the highest number of publications and collaborations, followed by China and Hong Kong. Asian countries, especially Hong Kong, have been active in exploring the use of large language models (LLMs) for essay scoring, especially in multilingual education contexts.

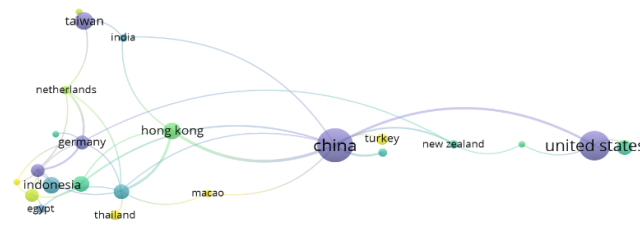


Figure 4. Co-authorship by country map



Figure 5. Co-authorship by organization map

3.4. Highly Cited Works

The most cited papers in this field focus on the ethics and reliability of AI in education. One of the top-cited articles is “Academic Integrity Considerations of AI Large Language Models in the Post-Pandemic Era: ChatGPT and Beyond”, which discusses fairness and academic honesty when using AI tools. Older studies about the e-rater® system and early AES algorithms remain important references for current researchers.

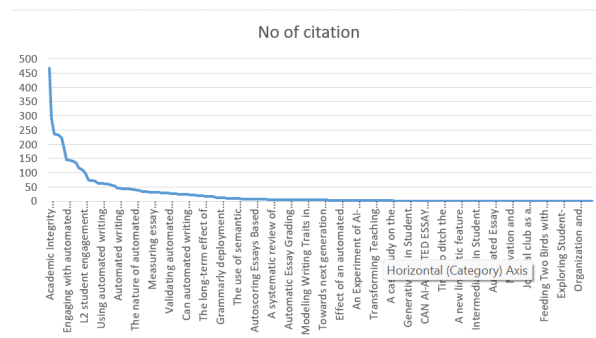


Figure 6. Top Citation

3.5. Main Research Themes

The keyword analysis identified several key themes in Automated Essay Scoring (AES) research. The first theme, AI and Feedback Systems, focuses on how AI tools such as Grammarly and other intelligent writing assistants enhance feedback and support student writing improvement. The second theme, Technical Foundations, encompasses studies involving natural language processing (NLP), machine learning, and deep learning methods used in developing AES models. The third theme, Education and Learning, highlights research that positions AES as a formative assessment tool aimed at improving learning outcomes rather than merely automating grading. The fourth theme, Language Learning, emphasizes the application of AES among non-native English learners (EFL/ESL) and in multilingual educational contexts. Over time, AES research has evolved from emphasizing technical accuracy to exploring its potential in enhancing learning and feedback practices, with recent studies examining the integration of ChatGPT and other generative AI tools for essay evaluation and intelligent feedback generation.

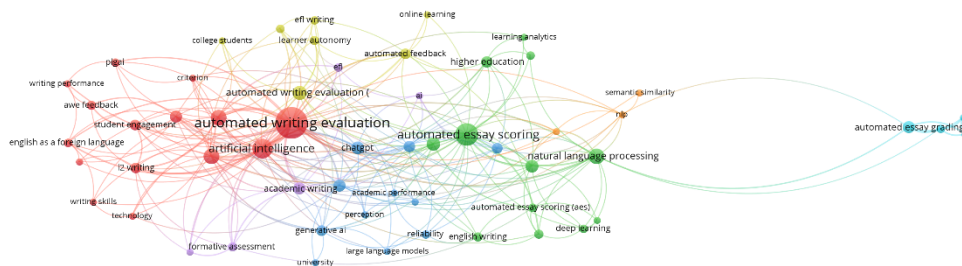


Figure 7. Indexed Keyword Co-occurrence Analysis

4. CONCLUSION

From 2010 to 2025, research on Automated Essay Scoring (AES) has expanded considerably, progressing from simple rule-based systems to sophisticated AI-powered models. The field has become increasingly global and interdisciplinary, integrating insights from education, linguistics, and computer science. Despite this progress, several areas still require further exploration. Future studies could focus on the use of generative AI to provide personalized writing feedback rather than just numerical scores, as well as enhancing fairness and explainability to ensure transparency and reduce bias in AES systems. Additionally, multilingual evaluation remains an important direction to test AES tools across diverse languages and cultural contexts, while long-term studies are needed to assess the impact of AES on student motivation, learning outcomes, and writing proficiency over time. In conclusion, as AI continues to advance, AES research must evolve beyond accuracy alone to ensure that automated assessment tools are fair, ethical, and genuinely supportive of learning in real-world educational environments.

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COMPARATIVE ANALYSIS OF KNOWLEDGE MANAGEMENT PRACTICES AND CHALLENGES: AN OBSERVATIONAL STUDY OF TWO SMALL AND MEDIUM ENTERPRISES (SMEs)

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ABSTRACT Despite the lack of recognized importance and practical application, knowledge management (KM) is inevitably a strategic capability for innovation and competitiveness in Small and Medium Enterprises (SMEs). However, many SMEs face persistent challenges in formalizing KM due to limited resources, informal work structures, and a lack of understanding of how knowledge processes unfold in daily operations. Consequently, KM in SMEs often suffers from fragmented handling of unstructured data, limiting the efficiency of knowledge acquisition, information retrieval, and storage processes. This study examines how two SMEs manage these foundational KM components through structured non-participant observation, addressing the lack of behavior-based research in this domain. Guided by the Knowledge-Based View (KBV) and Practice-Based View (PBV), a comparative multiple-case design was applied over eight weeks, focusing on daily data-handling practices, documentation routines, and retrieval activities. Findings reveal that both SMEs rely on unstructured data sources, yet differ in organization in terms of handling both explicit and tacit knowledge. In both cases, inconsistent storage and retrieval hindered knowledge accessibility and limited structured sharing. The study contributes theoretically by linking observable data practices to KM capability and methodologically by validating observation as a rigorous qualitative tool. Practical implications emphasize the need for SMEs to standardize digital storage and retrieval systems to enhance knowledge flow and organizational learning.

Keywords: knowledge management; small and medium enterprise; Knowledge-Based View; Practice-based View; knowledge acquisition; information retrieval.

1. INTRODUCTION

Effective knowledge management (KM) enables small and medium enterprises (SMEs) to transform unstructured data into valuable organizational knowledge. However, SMEs often lack systematic approaches for knowledge acquisition, information retrieval, and knowledge storage, resulting in fragmented repositories and inefficient access to information. While knowledge sharing remains essential, it is often shaped by the quality and organization of prior knowledge processes.

This study investigates how two SMEs manage unstructured data and transform it into usable knowledge through observable acquisition, retrieval, and storage practices. By emphasizing behavioral and process-based observation, the study addresses a key methodological gap in KM research, which typically relies on self-reported data.

The study aims to examine how unstructured data is managed, stored, and retrieved in SMEs, by comparing organizational routines related to knowledge acquisition and storage. In doing so, this study explores how these foundational processes influence knowledge sharing.

The research extends the theory Effective Personal Knowledge Management (PKM) [1] by highlighting the micro-level enactment of KM processes in real work contexts.

1.1. Theoretical Background

The Effective PKM framework conceptualizes how knowledge is managed at personal level, in which it contributes to the achievement of personal key performance index (KPI), and collectively the personal KPIs in an organization would contribute to the achievement of the organizational KPI [1, 2]. Derived and related to the fundamental theories of the SECI Model [3], the Effective PKM relates each KM process of socialization, externalization, combination, and internalization within the individual knowledge workers and the way they interact with the digital tools.

In understanding the concept of the use of digital tools in SMEs, it is inevitable to relate the technology with the process and the people within the organization, hence the need to consider the People-Process-Technology PPT [4] framework. This framework is used as a guide in analyzing the observation notes in this study. Covering the three elements in the PPT framework allows researchers to cover all grounds and evaluate the relationship between them in the context of the case studies.

Although there are other theories that conceptualize knowledge as critical resource [5] and emphasize that knowledge is enacted through daily routines and situated practices [6], only the integration of those theories is considered in this study, as the perspectives allows an examination of the observable mechanisms by which SMEs acquire, organize, and retrieve knowledge from unstructured sources.

The key KM processes relevant to this study include knowledge acquisition, information retrieval, and knowledge storage and organization. Knowledge acquisition involves identifying and capturing new information from internal and external sources. Information retrieval involves locating and accessing stored data for reuse in operations or decision-making. Knowledge storage and organization involve structuring and maintaining repositories, whether digital or physical. From these fundamental theories, an observation guide is devised emphasizing on the early stages of knowledge processing that influence subsequent sharing and innovation.

2. METHODOLOGY

A comparative multiple-case study approach was employed, focusing on two service provider SMEs:

- i. SME A: A technology-based service provider with semi-digital KM systems.
- ii. SME B: An education-based SME relying largely on manual and experiential methods.

Data were collected primarily through structured non-participant observation over an eight-week period, focusing on: data handling and documentation routines; information search and retrieval activities; storage, archiving, and version-control practices.

Field notes, document reviews, and internal reports were triangulated. Data analysis was conducted using thematic coding, guided by the KM process framework. Trustworthiness was maintained through prolonged engagement, reflexive journaling, and peer debriefing.

The overall process of this study consists of six steps:

- i. *Object and scope definition*: The objective of the comparative analysis is to identify the similarities and differences between the two selected case SMEs, with the scope of understanding how and why KM practices vary, exploring the contextual factors that shape these variations, and developing theoretical insights, extending relevant theories.
- ii. *Unit of comparison identification*: In identifying what is exactly being compared, this study takes into account the KM processes observed within the case SMEs: Knowledge acquisition (how knowledge is captured from unstructured data), information retrieval (how staff access stored knowledge), knowledge storage and organization (how data and knowledge are archived and maintained), and knowledge sharing (how retrieval and storage enable or hinder sharing).
- iii. *Analytical framework development*: A comparative framework is established in a form of a matrix that structures the analysis, consisting of columns (i.e., analytical dimension, SME code names, comparison insight), and rows (knowledge sources, knowledge acquisition methods, storage system, information retrieval, sharing mechanism).
- iv. *Thematic coding*: A procedure of open coding is performed to identify recurring behaviors, tools, and interactions, related to data handling in each theme/dimension. As much as possible, similar codes are grouped together under each KM dimension (axial coding).
- v. *Cross-case comparison*: Patterns across the two case SMEs are compared on each dimension, in the aspects of similarities, differences, and causal conditions (why).
- vi. *Comparative insights derivation*: Similarities and differences are translated into theoretical insights, transforming description into conceptual contribution.

3. FINDINGS AND DISCUSSION

Findings are discussed according to the key components of KM found relevant and necessary to the SMEs in study: knowledge acquisition and unstructured data management; information retrieval practices; knowledge storage and organization; and influence on knowledge sharing.

In general, Table 1 shows the findings on cross-case comparison between SME A and SME B, according to the analytical dimension based on KM Processes.

Table 1. Cross-case Comparison between the case SMEs

Analytical Dimension	SME A	SME B	Comparative Insights
Knowledge Sources	Unstructured, mainly documented reports (texts), diagrams (images), forms (texts) <i>Approximately 80% text data</i>	Unstructured, mainly documented reports (texts), statistical data in documents (texts), photos/graphics (images), videos <i>Approximately 40% text data</i>	The impact of unstructured data on KM practices in SMEs would be the increasing importance of preprocessing and interpretation, the rising role of contextualization and sensemaking, the changes in knowledge capture and storage models, the demands of new metrics and quality controls, and the increasing complexity despite the richer knowledge generation. <i>(DIKW Model)</i>
Knowledge Acquisition Methods	Search within folders, using keywords of file names, authors' initials	Search with folders, using keywords (only applicable for text data)	There is a need to form a mental model of where knowledge is located, select the right search strategies (keywords), evaluate hits, and iteratively refine queries. <i>(Search-as-Learning (SaL) Theory)</i>
Storage System	Internal centralized shared drive/cloud provided by proprietary platform (Microsoft suite) <i>Version control is applied on some data, yet still provide challenges especially to non-owners/non-creators of data.</i>	Internal public drive/cloud provided by commercial platform (Google suite) <i>Inconsistency view of data in storage, partly due to individual file/folder shared access by the owners/creators. Different type of data is stored in different locations.</i>	Need to analyze in the sense of provenance and traceability (what the memory records), retrievability and indexing (how memory is found), collaboration and social practices (who edits memory and how), consistency, conflict resolution and governance, and long-term preservation and archival. The theoretical impact to Knowledge Acquisition and KM practice includes memory richness versus accessibility trade-offs, design memory-aware workflows, hybrid architecture, and governance and metadata matter. <i>(Organizational Memory Theory)</i>
Information Retrieval	Log in access using staff ID, shared link to storage (as shared by data owner). <i>Complication when external contract-based staff is hired but does not have staff ID to access storage.</i>	Log in access using staff ID or shared ID, shared link to storage (as shared by data owner). <i>Complication when multiple platforms required multiple IDs, with some are shared IDs (e.g., for image data storage).</i>	The concepts of shared links versus canonical storage requires different perspectives, from "anyone with link" to policy-governed share links that embed attribute checks (who, when, on what device) and expiration. Logging and immutable audit trails should be enforced to maintain organizational memory and accountability. <i>(Zero Trust Architecture (ZTA) and Attribute-Based Access Control (ABAC))</i>
Sharing Mechanism	Shared link in email or open messaging platform (WhatsApp).	Shared files/data in open messaging platform (WhatsApp).	

Based on the tabulated matrix above, discussions could be derived in terms of knowledge dimensions or themes, i.e., knowing where, knowing who, knowing what, knowing how, and knowing when. *Knowing where* could be challenging (tacit knowledge) to most project members (staff), since it is

highly dependent on the human memory capacity. In most cases observed in SME A, due to the need for proper organization of data, the location of the data is moved, either by renaming the folder or moving the folder elsewhere, or moving the file/data to different folder, making the search and locating process difficult, especially when access permission needs to be reinstated to the changed folders. Meanwhile, in SME B, the challenge of knowing where is mainly on the abundant image data that are not properly organized with tags or labels, making it difficult to search, find, trace and recall.

Knowing who is basically the People factor, and it would be highly challenging for new member onboarding a project team. The recent practice in SME A is to name the file with author's initials to indicate who created or in charge of the data, but the person in charge may change upon the Project Director's instruction. In SME B, there is no trace of identity in file names. It is a norm in both SMEs to recall or ask in the messaging app (e.g., WhatsApp) to know who updated or owns the data. In most cases, it is assumed that the Managing Director in SME B has the final version of every data. Knowing who also applies to access verification over multiple devices for shared accounts.

Knowing what is observed as SME A maintained a central but loosely structured digital repository, while SME B's storage remained fragmented across personal devices and paper records, regardless the existence of some form of centralized storage. Both faced challenges in standardizing formats and ensuring version control. If the latest updated copy of confidential data is in an individual's personal hard disk, it is a challenge to know, search, request, and retrieve it.

Knowing how discussed here is based on the answer to "How to access, acquire and retrieve data?" Although it is clear in both SMEs that the central repository exists, the process is not well documented. Due to the dynamic situation, behavior, and intention of the data owners, and based on the understanding of their individual work process against management instruction, the process of accessing, acquiring, and retrieving data could be challenging for others. In SME B, it became a huge challenge when a project member unintentionally or forgot to save and share the updated file/data.

4. CONCLUSION

This study reveals that unstructured data management and the organization of knowledge repositories are pivotal yet often neglected in SMEs. Observation uncovered gaps between formal intent and actual practice, demonstrating that KM success depends on consistent knowledge acquisition and retrieval routines before sharing can be institutionalized. Future research should include integration of digital trace analytics or process mining, further enhance the understanding of unstructured data flows in KM systems.

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WEB-BASED PREDICTION OF REHABILITATION TRAINING EXERCISE FOR STROKE PATIENTS

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ABSTRACT Rehabilitation plays a vital role in restoring physical functionality and independence after illness or injury. Stroke, once predominantly affecting older adults, is increasingly impacting younger generations due to lifestyle factors such as obesity, hypertension, and diabetes. Limited time for exercise and poor dietary habits contribute to this growing risk. Early rehabilitation, especially within the first three to four months, is essential, and physiotherapists are crucial in guiding recovery through targeted exercises that improve strength, balance, and mobility. This study explores the prediction of rehabilitation exercises for stroke patients using machine learning algorithms. It evaluates the accuracy of Random Forest (RF), Multilayer Perceptron (MLP), and Logistic Regression (LR) based on a dataset of 5,110 stroke patient records. RF and LR both achieved 94.12% accuracy, while MLP slightly outperformed them with 94.32%. Despite this, RF was selected for deployment due to its stable prediction pattern and strong alignment with expert evaluations. The model was integrated into a web-based application hosted on Heroku, offering accessible rehabilitation guidance, particularly for patients in rural areas with limited access to professional care. The system's accuracy can be further improved through refined data collection and classifier tuning, making it a promising tool for personalised stroke recovery support.

Keywords: Machine Learning; Prediction Model; Rehabilitation; Stroke; Web Application.

1. INTRODUCTION

In the healthcare sector, data collection is crucial for generating a holistic view of improving communication between doctors and patients, personalising treatment, tracking genetic history, and advancing treatment methods [1,2]. The healthcare industry collects and records large amounts of data, but these data are not properly extracted or fully utilised. To date, several studies have been conducted that have focused on health prediction. These studies have applied different approaches to different problems and with different outcomes. One of the studies is on heart disease; experimental results showed an accurate classification accuracy of approximately 77% with a logistic-regression-derived discriminant function [3]. Other health diagnoses and predictions have been conducted year on year, categorised by critical diseases such as diabetes, heart failure, liver problems, cancer, mental illness, hepatitis, and stroke. Besides the value of healthy food and a refined diet, these need to be paralleled with physical exercise since the importance of regular physical activity in decreasing the risk of chronic disease is well established [4].

Exercise is categorised as a subcategory of physical activity which involves planned, structured, and repetitive movements for maintaining body health and condition. Regular physical activity and exercise help people to stay healthy, energetic, and play an essential role in preventing chronic diseases [5]. Regular exercise helps to build appropriate resistance in the muscles and prevent and reduce chronic diseases such as heart disease, obesity, blood sugar fluctuations, cardiovascular diseases, and cancer [6]. Cardiac rehabilitation exercises are developed and applied to improve the cardiovascular system for the prevention and rehabilitation of cardiac disorders and diseases. Exercise is not only used to improve health and maintain fitness but is also essential for physical rehabilitation [7-8]. For chronic diseases classified under modifiable risk factors, the treatment involves physical activity and exercise listed in the disease medical management plan [9]. However, inappropriate rehabilitation exercise could cause implications for the patient or lead to serious injury and, in the worst cases, may cause sudden death [10].

To assist physicians in the diagnosis process, decision support systems have been developed based on statistical data. The data might be outdated since the last examination was conducted, such as MRI, X-ray, chemotherapy, and other medical examinations. In the healthcare sector, decision support systems are also known as clinical decision support systems (CDSS) or medical decision support systems (MDSS). The purpose of CDSS is to improve healthcare delivery by enhancing medical decisions with specific clinical knowledge, patient information, and other health information (Sutton, 2020). CDSS is developed with the ability to leverage data and observations; thus professional physicians can combine their knowledge with information provided by CDSS [11]. A CDSS capable of learning the relations between patient history, population diseases, symptoms, test results, disease pathology, and family history is very valuable to professional physicians and healthcare institutions [12].

Diverse approaches are being used to develop health prediction systems, such as machine learning, fuzzy logic, rule-based systems, big data, and data mining. Each of the approaches and methods produces different results and evaluations. The contribution of health prediction in critical diseases helps the healthcare sector to provide an early diagnosis and suggest steps to be taken. However, most of the predictions conducted focus more on non-physical diagnosis and suggestions. There is still a lack of research and development on the prediction of rehabilitation exercises that can help healthcare centres with the rehabilitation process. Since there is a limitation in accessing rehabilitation facilities and private professional help for critical patients, especially in rural areas, the rehabilitation exercise system with health prediction and suggestions could help to minimise the gap.

To generate accurate predictions, the parameters need to be very specific and require a domain expert's help. Prediction requires a large amount of data analysis and rule statements created in the algorithm [13]. To generate the prediction, data need to be trained and undergo testing repeatedly. The problem is that prediction requires vast amounts of data to train the algorithm, and the accuracy is dependent on the consistency of the data training. For example, patients need to describe their current conditions, medical histories, and family histories so that web-based prediction can be more precise and specific for the patient [1]. The patient might not have a similar type of chronic disease, and a rehabilitation exercise recommendation may be generated in the prediction. Data collection parameters need to be more focused on the current patient condition and state.

Even though the learning process rate is faster than human processing, it can still be considered time-consuming and gradual. The prediction precision can be improved gradually by increasing the data training and data testing. As an example, each patient has a different medical condition; thus, the prediction result for rehabilitation exercise differs for each patient. The approaches are tailored to the previous health condition, and this suggests less precision and less focus on the recovery process. The problems highlighted below in the problem statement focus on healthcare prediction:

- i. Large amounts of unspecific patient data are recorded
- ii. The diagnosis and suggestions are very general
- iii. Lack of prediction in developing rehabilitation exercises for chronic diseases
- iv. The approaches are tailored to the previous patient's health condition.

This research helps patients to have effective rehabilitation outside the rehabilitation centre. The prediction of the rehabilitation exercise is able to help address the issue of limited medical professionals. This might be handy if it can be used in rural areas that have limited access to the rehabilitation centre. Rehabilitation exercise is recommended to be done regularly, especially for chronic disease patients, as an alternative treatment. However, inappropriate exercise may cause many other implications. Thus, this study aims to predict the rehabilitation exercises that can be done for chronic disease patients in rehabilitation sessions.

Several studies and web developments have been conducted in health prediction; however, there is still a lack of prediction studies and development on rehabilitation exercises. This study aids in selecting the best algorithm to be used to develop a web-based prediction system that can generate recommendations for rehabilitation exercises for chronic disease patients. This aids in finding the most precise prediction for the rehabilitation exercises that can be used in daily exercise at any fitness centre.

2. METHODOLOGY

The research methodology consists of data collection, pre-processing, modelling, evaluation, and web deployment (Figure 1).

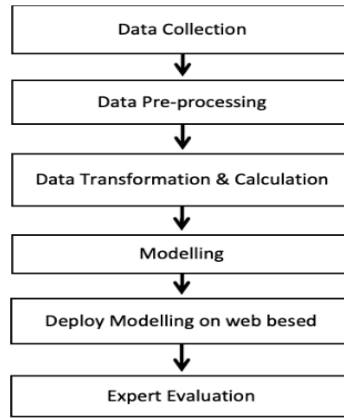


Figure 1. Methodology of the research

2.1 Data Collection

The dataset, obtained from an open-source stroke dataset containing 5,110 patient records, includes attributes such as age, gender, glucose level, body mass index (BMI), smoking status, paralysis type, and speech ability. These parameters were selected because they directly influence recovery progress and exercise suitability. Once the dataset was acquired, it underwent a comprehensive pre-processing phase to ensure data quality and consistency before model training.

2.2 Data Pre-Processing and Transformation

Raw data were cleaned to remove duplicates and missing values, reducing 8,017 records to 5,110 valid entries. Categorical values were converted into numeric form (e.g., male = 1, female = 0) for model training. Continuous data, such as BMI and glucose level, were normalised to reduce dimensionality and improve prediction accuracy. Patients were classified into health categories based on stroke severity and BMI range to balance the dataset. With the cleaned and transformed dataset prepared, the next phase involved implementing and comparing multiple machine learning algorithms to identify the most suitable model for predicting rehabilitation exercises.

2.3 Modelling

Three supervised learning algorithms were tested: Random Forest (RF), Logistic Regression (LR), and Multilayer Perceptron (MLP). The dataset was divided into 70% training and 30% testing using stratified sampling. RF performed most consistently, yielding an accuracy of 94.12%, compared to 94.32% for MLP and 94.12% for LR. Despite MLP's slightly higher accuracy, RF was selected due to its stability, interpretability, and alignment with expert assessments. Following the selection of the RF algorithm, a rigorous evaluation process was conducted to validate the model's performance and ensure its applicability in real-world clinical settings.

2.4 Model Evaluation

Cross-validation was performed to ensure generalisation, and evaluation metrics such as accuracy and error rate were computed. The prediction results were validated by two professional rehabilitation experts from a registered PERKESO rehabilitation centre. Expert assessment confirmed that over 80% of RF predictions matched expert recommendations, verifying its reliability for clinical application.

3. WEB-BASED APPLICATION DEVELOPMENT

The finalised RF model was integrated into a web-based application hosted on Heroku, enabling users to access the system remotely. All code was stored in a GitHub repository linked to Heroku for continuous deployment and version control.

The system interface consists of three main components:

- i. Model Interface: Displays data input forms for patient information.
- ii. Main Page: Provides general information about stroke rehabilitation.
- iii. Exercise Prediction Page: Generates predicted rehabilitation exercises based on patient input.

The system aims to assist patients who have limited access to professional rehabilitation centres, especially in rural communities. Through its prediction feature, users can obtain exercise suggestions aligned with their recovery level and physiological status.

4. TESTING AND EVALUATION

Testing was conducted to measure the accuracy and functionality of the web-based system. Functional testing verified system stability, while predictive testing measured the model's precision using [14] Equation (1):

$$E = \{E_i \mid i = 1, \dots, n\} \quad (1)$$

where E_i represents individual prediction errors.

Evaluation by rehabilitation specialists showed over 80% agreement with the system's predictions, confirming its reliability and potential for supporting real rehabilitation programmes.

The experts also suggested enhancing dataset diversity to better represent various stroke conditions and optimising classifier parameters to further improve prediction accuracy and system adaptability in future development.

5. CONCLUSION

This study developed a machine learning-based prediction model for stroke rehabilitation exercises. Using data from 5,110 patients, RF, LR, and MLP algorithms were compared. RF achieved the most stable performance (94.12% accuracy) and was selected for web deployment.

The resulting web-based prediction system, hosted on Heroku, provides accessible rehabilitation guidance, helping stroke patients engage in appropriate exercises outside clinical facilities. Expert evaluation confirmed the model's effectiveness and reliability. This study contributes to the growing body of evidence supporting machine learning applications in personalised healthcare, particularly in resource-limited settings where access to rehabilitation specialists remains challenging. Future work focuses on expanding datasets, tuning algorithms, and incorporating personalised monitoring features for adaptive rehabilitation recommendations.

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ADAPTIVE HUMAN-MACHINE TEAMING FRAMEWORK FOR EXPLAINABLE AND SUSTAINABLE STROKE MANAGEMENT DECISION SUPPORT

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ABSTRACT This paper presents an Adaptive Human-Machine Teaming framework for explainable and sustainable stroke management that integrates adaptive counterfactual reasoning, interactive mechanisms, and co-learning processes among clinicians, AI system administrators, and intelligent systems. Existing AI models for hemorrhagic transformation risk prediction remain largely algorithm-centric, offering static explainability and limited workflow integration. The proposed framework redefines decision support as a collaborative ecosystem rather than a one-way predictive pipeline. Its adaptive counterfactual module continuously refines “what-if” explanations using clinician feedback and real-time data updates, ensuring clinically plausible and context-sensitive insights. An interactive mechanism connects doctors, AI managers, and the system through shared dashboards and feedback channels, enabling transparent model adjustments, drift monitoring, and explainability governance. A co-learning mechanism closes the loop, where clinician corrections and administrator oversight dynamically retrain and calibrate the model through interactive machine learning. Together, these components transform traditional decision support systems into adaptive, trustworthy, and auditable clinical intelligence platforms, enhancing both decision-level reasoning and system-level accountability. The framework demonstrates how adaptive counterfactuals and dual-user collaboration can advance AI reliability, interpretability, and sustainability in real-world stroke care.

Keywords: Acute ischemic stroke; hemorrhagic transformation; explainable AI; radiomics; counterfactual reasoning; decision support system

1. INTRODUCTION

Hemorrhagic transformation (HT) following acute ischemic stroke (AIS) remains a major complication of intravenous thrombolysis, often resulting in increased morbidity and mortality. Early prediction of HT risk is crucial for guiding safer and more effective treatment. Although recent artificial intelligence (AI) models have achieved promising predictive performance [1, 2], most remain algorithm-centric, emphasizing accuracy over clinical integration, adaptability, and explainability. These systems function as stand-alone predictors, providing static outputs that fail to engage clinicians in reasoning or reflect the dynamics of real-world workflows.

Explainable AI (XAI) [3] and counterfactual [4] reasoning have been introduced to improve interpretability through visual explanations and “what-if” analyses. However, existing counterfactuals are typically static [5], which utilises predefined scenarios that do not learn from user input or update as patient data evolve. Moreover, current systems focus almost exclusively on clinician-facing dashboards [6, 7], neglecting the critical role of AI system administrators who ensure model maintenance, governance, and explainability stability. Without structured feedback from clinicians and oversight from administrators, deployed AI systems lose reliability and transparency over time.

To overcome these limitations, this study proposes an Adaptive Human-Machine Teaming (HMT) framework for explainable stroke management that integrates three interrelated components: adaptive counterfactual reasoning, interactive mechanisms, and co-learning processes among doctors, AI managers, and the intelligent system. The adaptive counterfactual module continuously refines its “what-if” explanations through clinician feedback using counterfactual active learning [8] and interactive surrogate editing [9], ensuring more context-sensitive and clinically plausible reasoning. The

interactive mechanism establishes a dual-user environment where clinicians co-explore predictions and explanations, while AI administrators monitor drift, retraining triggers, and explainability metrics through secure, auditable dashboards [10]. Finally, the co-learning mechanism closes the feedback loop via interactive and online learning, where clinician corrections and administrator oversight guide incremental model updates and recalibration.

By combining these elements, the Adaptive HMT framework transforms static, one-way prediction systems into collaborative, continuously learning decision ecosystems. It bridges decision-level reasoning and system-level governance, ensuring that AI models in stroke care remain interpretable, adaptive, and trustworthy across their lifecycle.

2. MATERIALS AND METHODS

The proposed Adaptive HMT framework (*Figure 1*) is currently at the requirement evaluation stage, involving clinicians and AI system administrators as pilot stakeholders. Designed to address persistent limitations in existing DSS, the framework introduces several key innovations. It incorporates an adaptive counterfactual generation module, a capability not found in current stroke decision support systems, which uses counterfactual active learning and interactive surrogate editing to continuously refine explanations based on clinician interactions and real-time data updates. An interactive model maintenance loop links user feedback directly to model retraining triggers through a combination of coactive learning and online learning, enabling continuous improvement in both predictive performance and interpretability.

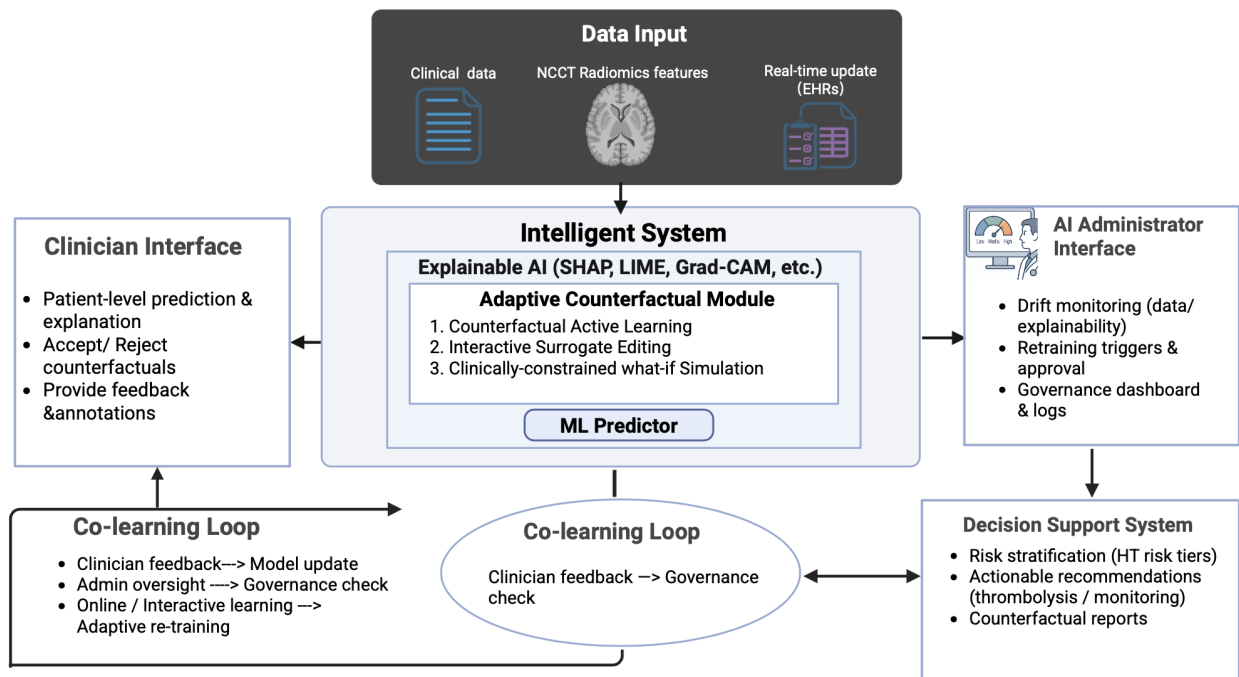


Figure 1. Proposed Adaptive Human Machine Teaming-driven Framework for Stroke Management

The proposed adaptive HMT dashboard is designed to support two key user groups: AI system administrators and clinicians. While clinicians use the dashboard for patient-level decision support, administrators oversee model reliability, governance, and lifecycle maintenance. Both users interact with the same intelligent system, but their interfaces, explanation depth, and adaptive functions differ according to their respective roles and responsibilities.

The first component, adaptive counterfactual reasoning, introduces a feedback-driven approach where “what-if” explanations evolve dynamically based on clinician interactions and patient data updates. By employing counterfactual active learning and interactive surrogate editing, the system learns from

accepted or rejected scenarios, continuously refining its causal understanding to produce context-sensitive and clinically plausible explanations.

The second component establishes an interactive mechanism that connects doctors, AI managers, and the intelligent system within a unified dashboard environment. Clinicians interact through adaptive visualizations and explainable interfaces to explore, validate, and refine AI predictions. Simultaneously, administrators monitor explainability drift, data drift, and model performance, approving or scheduling retraining as necessary. This interaction layer ensures transparency and traceability, creating an accountable governance structure for both human and machine actions.

The third component introduces a co-learning mechanism in which clinician feedback and administrator oversight are used to incrementally retrain or recalibrate the model through interactive machine learning and online learning algorithms. The AI learns from human corrections, adjusts its prediction boundaries, and reconfigures its counterfactual logic, while administrators ensure the stability and safety of each update. This dual-loop learning process enables continuous adaptation which aligns algorithmic reasoning with clinical expertise and system governance.

3. CONCLUSION

This research is firmly situated within the data science domain, as it integrates data-driven modeling, explainable analytics, and human-centered feedback into a unified decision support framework. The proposed adaptive HMT framework employs core data science methodologies with machine learning for prediction, explainable and counterfactual AI for interpretability, and online or coactive learning for continuous model improvement.

Beyond algorithmic performance, the work addresses how data, models, and human expertise interact dynamically through adaptive dashboards and co-learning mechanisms. This positions the study within the emerging frontier of interactive and lifecycle-aware data science, where the emphasis extends from generating accurate predictions to ensuring that insights remain explainable, adaptive, and trustworthy throughout deployment.

The framework further advances a dual-layer human-machine teaming architecture, extending beyond clinician-only interaction to include AI system administrators as active collaborators who monitor model adaptivity, drift, and explainability stability. In addition, it features an adaptive dashboard architecture that dynamically adjusts visualization depth, explanation granularity, and interaction modes according to user role and clinical workflow context. Finally, the framework embeds governance-aware explainability, systematically monitoring explainability drift and transparency consistency across model versions to ensure sustained reliability, accountability, and trustworthiness throughout the AI lifecycle.

By bridging technical modeling with user collaboration and governance, the framework advances data science toward sustainable, human-in-the-loop intelligence systems for clinical decision support.

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REAL ESTATE ANALYTICS AND PRICE PREDICTION USING AUTOMATED MACHINE LEARNING

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ABSTRACT The Malaysian real estate market is characterized by a dynamic nature and a variety of factors that influences the prices of the real estate, which makes it quite challenging in providing precise real estate price predictions. The current research meets the demand in the field of data driven decision making tools by creating an analytics and price prediction system of real estate with the help of machine learning algorithms, focusing on Selangor, Kuala Lumpur, Penang, and Johor. Using PyCaret, an Automated Machine Learning (AutoML) tool, system automatically pre-processes data, selects and configures model hyperparameters to find the best predictive models. Different regression models were considered and optimized on performance parameters that are, Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 . The best model was implemented as an interactive web-based dashboard which provides stakeholders with price predictions and market information. This project proves that AutoML is a viable option to simplify the complex machine learning processes and offers a robust and user-friendly product to predict real estate prices in the Malaysian context in accurately and efficiently. Future work involves extracting and analyzing data from other countries in Malaysia and other AutoML tools.

Keywords: AutoML, house price prediction; machine learning; PyCaret.

1. INTRODUCTION

The Malaysian real estate market is highly dynamic, influenced by multiple factors such as location, property type, and economic conditions, making accurate price prediction challenging [1]. Traditional valuation methods are often time-consuming and prone to bias. Machine learning (ML) offers a data-driven approach capable of handling large datasets and capturing complex relationships [2]. Despite global advancements, studies focusing on Malaysia remain limited. This project aims to develop an automated ML-based real estate price prediction system for Selangor, Kuala Lumpur, Penang, and Johor, leveraging PyCaret to simplify model development and deployment [3].

2. MATERIALS AND METHODS

The real estate price predictions are based on historical transaction data for Selangor, Kuala Lumpur, Penang and Johor from January 2021 to June 2024, that consists of data regarding property type, district, 'mukim', area, date of transaction, land area, tenure, and transaction price [4]. Preprocessing on the data involved converting the text data into categorical data, delete rows with missing data, irregular values, and duplicate data, normalization, and remove outliers. PyCaret's AutoML tool was employed to automate model selection, hyperparameter tuning, and evaluation. Multiple regression algorithms—Random Forest, Extra Trees, XGBoost, LightGBM—were compared using 10-fold cross-validation. Performance metrics included Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 . The best-performing model was integrated into a Streamlit-based dashboard for real-time predictions and market trend visualization [5].

3. RESULTS AND FINDINGS

Among all tested models, ensemble methods outperformed others. The Extra Trees Regressor achieved the highest accuracy across combined regional datasets, with $R^2 \gg 0.903$, MAE $\gg$ RM49,753, and RMSE $\gg$ RM79,161 after fine-tuning [6]. Region-specific models also showed strong performance, with Selangor and Penang achieving $R^2 > 0.91$. Visual analysis confirmed good predictive reliability, though slight heteroscedasticity was observed at high price ranges. The deployed dashboard, PropVision Malaysia, provides interactive features for property price prediction, market trend analysis, and model insights, enabling stakeholders to make informed decisions [3].

4. CONCLUSION

This study demonstrates the feasibility of using AutoML for real estate price prediction in Malaysia. PyCaret significantly reduced development time while ensuring robust model performance. The analysis was based on applying a structured data preprocessing pipeline to four regional datasets of Selangor, Kuala Lumpur, Penang, and Johor. The performances of the models were evaluated in terms of R^2 , RMSE, and MAE in regional and all the four-country datasets. Extra Trees Regressor produced the most stable and accurate results with high accuracy and minimal prediction errors, and this same model can be used for accurate price prediction for all four countries. Future research involves extracting and analysing price prediction from other countries in Malaysia, applying PyCaret and other AutoML tools.

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RISK STRATIFICATION MODELLING OF INAPPROPRIATE NON-STEROIDAL ANTI-INFLAMMATORY DRUGS (NSAIDS) USE FOR OSTEOARTHRITIS USING ARTIFICIAL NEURAL NETWORKS AND MODEL VALIDATION IN HIGH-RISK COMMUNITY-DWELLING OLDER ADULTS

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ABSTRACT Osteoarthritis (OA) is a major cause of disability among older adults, with non-steroidal anti-inflammatory drugs (NSAIDs) commonly used for symptom relief despite their known risk of adverse drug reactions (ADRs). Older adults are particularly vulnerable due to age-related frailty, comorbidities, and polypharmacy. In Malaysia, NSAID-related ADRs remain a significant contributor to medication harm based on national pharmacovigilance data. Current prescribing risk tools are limited by static scoring systems and lack of local validation. To develop and validate machine learning-based risk prediction model for inappropriate NSAID use among older adults using real-world pharmacovigilance data. Logistic Regression (LASSO), Random Forest, XGBoost, and Artificial Neural Network (ANN) models will be trained and validated using k-fold cross-validation to identify high-risk individuals before ADRs occur. Model performance will be evaluated using AUROC, sensitivity, specificity, and calibration metrics. The machine learning model to accurately identify high-risk older adults before ADRs occur, enabling proactive interventions.

Keywords: NSAIDs; Adverse drug reactions; Osteoarthritis; Machine learning; Older adults

1. INTRODUCTION

Osteoarthritis (OA) has now become one of the leading diseases that cause disability among older people and in recent years, the world sees approximately 7.6% of its population is affected by OA. The Global Burden of Disease (2019) reported that numbers of incidence, prevalence and loss of healthy life are increasing over the years, particularly among aging populations. Although OA does not significantly contribute to the mortality burden as compared to cardiovascular diseases and cancer, it remarkably compromises the patients' quality of life and necessitates healthcare services. [1, 2] While researchers focus on the association between OA and age-related physiological decline (frailty) and developing effective treatments to at least slow down OA degeneration process, patients mainly treated with anti-inflammatory and pain killers to alleviate symptoms. This includes non-steroidal anti-inflammatory drugs (NSAIDs) such as celecoxib and diclofenac, usually paired with paracetamol, glucocorticoids or opioids. [3, 4, 5] However, widely prescribed NSAIDs could carry risk of serious Adverse Drug Reactions (ADRs).

Older adults are at higher risk due to frailty, comorbidities and polypharmacy. As this age group grows year by year, particularly ones with OA, more national budget need to be allocated, signify increased healthcare needs for this population. [6, 7] It has been reported that NSAIDs contributed up to 30% of ADR hospitalization due to cardiovascular, gastrointestinal, and renal complications. [8] Studies have shown that some hospital admissions are preventable with appropriate medication prescribing to minimize risk of have side effects and ADR, whilst improving both monitoring and adherence. In Malaysia, analgesics including NSAIDs ranked among the top causes of ADR reports based on the pharmacovigilance data from National Pharmaceutical Regulatory Agency (NPRA). Therefore, AI-based risk stratification model would be valuable to support and validate subjective clinical judgment and proactively identify high-risk individuals before harm occurs. [9, 10, 11, 12, 13, 14]

Existing NSAID prescribing risk tools are not AI-driven and rely on static clinical judgment. To date, there is no predictive model validated specifically for Malaysian older adults using real-world NPRA pharmacovigilance data. Most prior models do not integrate multi-factor risk profiling (e.g., demographic, polypharmacy, comorbidity, prescriber setting). Therefore, machine learning-based risk stratification model should be developed or clinically evaluated in this population to enable proactive early intervention. This research aim to develop and validate machine learning-based risk prediction model using NPRA pharmacovigilance data.

2. METHODS

- Study design: Retrospective pharmacovigilance-based predictive modelling study.
- Data source: Adverse Drug Reaction (ADR) reports involving adults ≥ 60 years extracted from the NPRA database.
- Outcome: Inappropriate NSAID-related ADRs (as defined by NPRA MADRAC coding).
- Predictor variables: Age, gender, ethnicity, body weight, comorbidities, polypharmacy, drug class, concomitant medications, prescriber setting.
- AI model: Logistic Regression (LASSO), Random Forest, XGBoost, Artificial Neural Network (ANN) trained using 70:30 train-test split and validated using k-fold cross-validation.
- Performance metrics: AUROC, sensitivity, specificity, and calibration score.

3. EXPECTED OUTCOME

We expect the machine learning model to accurately identify high-risk older adults before ADRs occur, enabling proactive interventions.

4. CONCLUSION

This study aims to enable early identification of older adults at high risk of NSAID-related adverse drug reactions through a proactive, AI-driven approach to patient safety. The proposed model represents the first Malaysia-specific machine learning tool developed using NPRA pharmacovigilance data, supporting precision prescribing in both primary care and community settings. By facilitating safer, data-informed decisions, the model has the potential to reduce preventable hospital admissions, healthcare costs, and morbidity associated with inappropriate NSAID use. Moreover, this framework can be extended to other high-risk medications in the future, further strengthening medication safety in Malaysia.

ACKNOWLEDGEMENT

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TEXTUAL ADVERSARIAL EXAMPLE GENERATION SYSTEM

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ABSTRACT- The generation of textual adversarial examples is vital for enhancing the robustness of natural language processing (NLP) models, particularly in sentiment analysis. To address this challenge, the study presents a system designed to automate the creation of high-quality adversarial examples using the BU-SPOF method. This project focuses on preserving semantic consistency, grammatical accuracy, and lexical correctness while generating adversarial texts. The approach utilizes a dataset of annotated sentiment samples, which are analyzed to construct a web-based application that systematically generates linguistically coherent adversarial examples.

Keywords: Adversarial examples, sentiment analysis, semantic consistency, text perturbations.

1. INTRODUCTION

In the rapidly evolving field of natural language processing (NLP), textual adversarial examples play a crucial role in driving advancements, particularly in sentiment analysis. These examples are instrumental in revealing vulnerabilities within NLP models, as evidenced by state-of-the-art adversarial attacks that effectively induce errors in model outputs [1]. Employing adversarial training techniques, textual adversarial examples are essential for enhancing the robustness and resilience of sentiment analysis models. Through deliberate exposure to crafted adversarial inputs, sentiment analysis models can develop a heightened ability to navigate complex linguistic nuances and potential adversarial challenges, contributing to their improved performance and adaptability in real-world applications. This underscores the crucial role of textual adversarial examples in shaping the evolution and efficacy of sentiment analysis within the dynamic landscape of NLP.

Textual adversarial examples are crafted inputs that mislead machine learning models in natural language processing (NLP) by making a few subtle perturbations to the original text. These perturbations are categorized into three levels: character-level, word-level, and sentence-level [3]. Character-level attacks modify individual characters through changes like swapping or substitution, aiming for undetectable alterations that impact model predictions [4]. Word-level perturbations maintain semantic and syntactic integrity by replacing words with synonyms or typos, while ensuring text coherence [5]. Sentence-level perturbations involve more significant modifications to entire sentences, altering the overall context or narrative, although they can sometimes produce text that is unreadable [6]. Each level of perturbation targets different model vulnerabilities and addresses the complexities of language.

Sentiment analysis is a branch of natural language processing (NLP) focused on identifying the emotional tone within text. It classifies sentiments as positive, negative, or neutral and employs methods like lexicon-based systems, machine learning, and hybrid approaches [7]. Widely used for social media monitoring, customer feedback analysis, and brand reputation management, sentiment analysis helps businesses and researchers glean insights into public opinions, market trends, and sentiment around topics or brands, providing a deeper understanding of emotional responses in written content.

The BU-SPO (Bigram Unigram-Semantic Preservation Optimization) algorithm is designed to create semantically coherent textual adversarial examples, specifically targeting black-box word-level attacks where model parameters and gradient information are inaccessible [8]. Unlike traditional unigram-based attacks, BU-SPO uses both unigram and bigram modifications, enhancing the authenticity of adversarial text by replacing entire bigrams with semantically similar phrases (e.g., changing "mad doctor" to "mental doctor").

2. MATERIALS AND METHODS

This section outlines the stages in developing a web-based system for generating textual adversarial examples. The process begins with the construction of the web application, establishing the user interface and foundational structure. Once the application is set up, the algorithm is integrated to enable the generation of adversarial examples. Following this, a sentiment analysis model is implemented to examine the accuracy of the generated textual adversarial examples.

2.1. Generating Adversarial Examples

Figure 1 outlines the steps for generating textual adversarial examples. The initiation involves the user choosing between two input types: A and B. For type A, the user inputs their own text, while for type B, the user selects a dataset and picks a sample text. Once the input is provided, the user can choose to generate adversarial examples. Upon selection, the generated adversarial example is shown on the screen and simultaneously stored in the database.

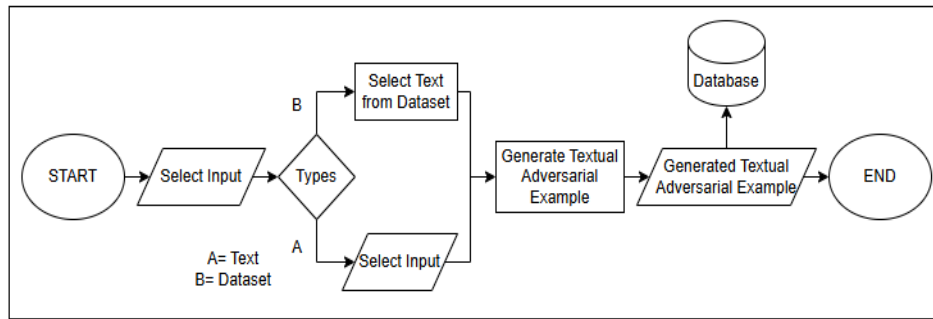


Figure 1. Steps to generate textual adversarial examples

The *difference in vector representation* measures the change in the numerical representation of the text before and after perturbation. Specifically, it assesses the difference in the vectorized form of the original text compared to its altered counterpart. The purpose of this measurement is to determine how significantly adversarial perturbations have altered the text's representation. A larger difference typically indicates that the adversarial modifications have had a more substantial impact on the text's vector, which can influence the model's predictions. This is important because more significant changes in vector representation suggest that the adversarial text is more likely to affect the model's behavior.

The *probability shift* measure focuses on the change in the probability distribution produced by a model, for the original sentiment class after the text has been perturbed. It calculates the difference between the probability assigned to the original class and the probability assigned after the perturbation. A larger probability shift signifies that the adversarial text has successfully altered the model's confidence, making it a more effective adversarial example. This shift indicates how well the perturbations have fooled the model into reconsidering its classification, which is crucial for assessing the strength of the adversarial text.

3. RESULTS / FINDINGS

The *probability shift* shows the influence of the adversarial modifications towards a classifier's predictions. Additionally, the Universal Sentence Encoder (USE) score evaluates semantic similarity, indicating the extent of meaning alteration. Each adversarial text's status, marked as a success or failure, reveals whether it effectively fooled the classifier. This analysis provides a comprehensive view of the adversarial attack's success and its influence on classification accuracy.

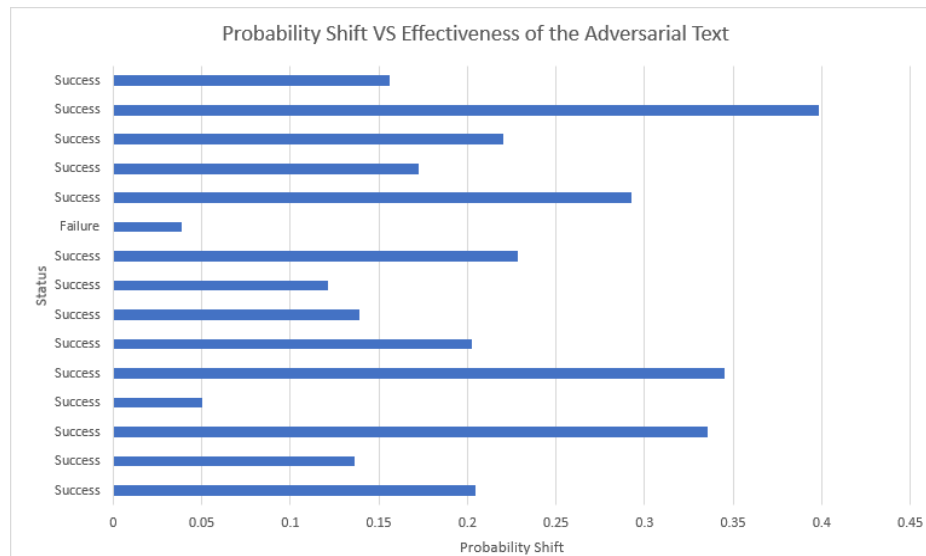


Figure 2. Chart on the Relationship between the Probability Shift and the Status of the Adversarial Texts

In our analysis, successful adversarial texts demonstrate Probability Shifts from about 0.0387 to 0.3985, suggesting that higher shifts are generally linked with success. In contrast, the single failed instance shows a low *probability shift* of 0.0387 and a low USE score, which indicates that the minimal probability change may reduce the likelihood of a successful adversarial impact.

This data suggests that *probability shifts* under roughly 0.1 may decrease the effectiveness of the adversarial texts. In addition, the successful examples tend to show higher Probability Shifts, often critical in achieving the intended adversarial outcomes.

4. CONCLUSION

The application assesses the impact of these examples on sentiment predictions, providing valuable insights into their effectiveness. By improving the adaptability and resilience of sentiment analysis models, this project aims to contribute significantly to the field of NLP, ensuring models can effectively navigate real-world linguistic complexities.

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STRATEGIES OPTIMIZATION FOR RUGBY TEAM USING GENETIC ALGORITHM (GA)

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ABSTRACT This paper provides a literature review on the application of Genetic Algorithms (GA) to improve strategic decision making for rugby teams. This review explores that beyond traditional reliance on coach intuition and experience. The primary goal is to support coaches and analyst in efficiently generating and evaluating the most effective tactical for match day. GA is an evolutionary algorithm that is well-suited for solving complex problems with a large number of possible solutions, which is essential given the many combinations of tactical plays in rugby. This review finds that the conceptual framework involves representing a complete rugby strategy as a chromosome, with individual tactical decisions (e.g., pass, run, kick) encoded as genes. The core GA process are involving initialization, fitness evaluation, selection, crossover, and mutation. It is employed to iteratively evolve strategies towards an optimal outcome. Strategy quality is measured using a fitness function that integrates predefined Key Performance Indicators (KPIs) from studies in Rugby Union and Rugby Sevens, such as maximizing tries scored and restarts won, while minimizing penalties and missed tackles. By implementing this data-driven approach, the system aims to identify and implement more effective strategies, thereby improving team performance and achieving a competitive advantage

Keywords: Genetic Algorithm (GA); Optimization; Rugby Strategy; Key Performance Indicators (KPIs); Decision-Making

1. INTRODUCTION

This article provides a literature review on the application of Genetic Algorithms (GA) for optimizing rugby team strategies. This review examines the existing literature to identify key components which are successful applications in other sports and the current research gap in the context of rugby. Rugby is an increasingly popular sport in Malaysia, with over 300 clubs and 600 schools actively involved [1]. However, strategy planning in rugby still faces challenges. Most coaches rely on intuition and experience, without systematic data support tools for decision-making [2]. This reliance on conventional tactics makes teams predictable to opponents. Although Genetic Algorithm (GA) has been successfully used in other sports like basketball to optimize training and game tactics [3], or in rugby for simpler physical aspects like optimizing kick angles [4], its application for dynamic and complex team strategy optimization remains unexplored. A research gap exists in developing a GA model that uses rugby-specific indicators like position, player strength, player weaknesses and player attributes. Therefore, this literature review aims to combine existing research to establish a foundation for a future optimization system in rugby teams. This paper will use real-world indicators identified from the literature to recommend the most effective strategy combinations.

2. MATERIALS AND METHODS

The methodology involves a literature review and the conceptual design of an optimization system based on a Genetic Algorithm (GA). Optimization serves as a practical technique for addressing complex decision-making problems, particularly in fields that demand strategic planning such as sports. For instance, rugby coaches often face difficulties in creating effective game strategies and tend to rely heavily on previous experience and intuition, as noted in [1]. To enhance this process, optimization methods such as the Genetic Algorithm (GA) have been introduced. A study in [3] demonstrated the success of GA in refining both the training schedules and tactical decisions of basketball players. Optimization tools are best used to support strategic decision-making. According to [5], optimization

involves identifying the best or most efficient solution from all available options. Therefore, optimization tools should not be used to replace the decision-making process, but they should be employed to support it [5]. This prototype will use Genetic Algorithm (GA), which is a type of evolutionary algorithm based on the mechanism of biological evolution. The overall conceptual framework of the proposed system is illustrated in Fig. 1. The process begins with the user (coach/analyst) interacting through the user interface (UI). The processed data is then fed into the Genetic Algorithm (GA) module, which consists of several stages: initialization, fitness evaluation, selection, crossover, mutation, and termination. Finally, the system produces optimized strategies as output.

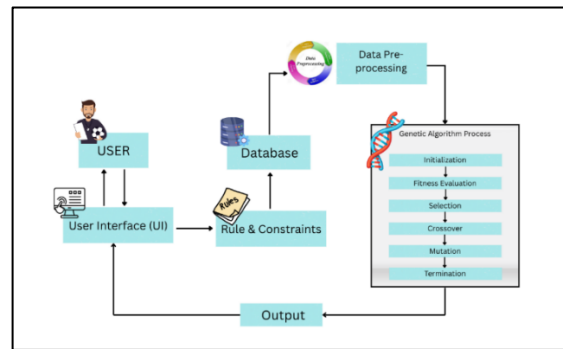


Figure 1. System Architecture

The overall conceptual framework of the proposed system is illustrated in Fig. 1. The process begins with the user (coach/analyst) interacting through the user interface (UI). The processed data is then fed into the Genetic Algorithm (GA) module, which consists of several stages: initialization, fitness evaluation, selection, crossover, mutation, and termination. Finally, the system produces optimized strategies as output.

In this prototype, each chromosome corresponds to a complete rugby strategy, while each gene represents a single tactical decision or play (e.g., pass, run, or kick), as also described in [6]. The main components are:

Population Initialization: The process begins with the creation of an initial population of possible solutions, represented as chromosomes.

Chromosome Evaluation: Each chromosome is evaluated using a fitness function, which measures the quality of the solution. This assesses how close a solution is to the optimal outcome, as demonstrated in [6]. In this project, the fitness function uses Key Performance Indicators (KPIs), rewarding strategies that lead to more tries or penalizing those with high missed tackle counts.

Selection: This involves choosing the chromosomes with the highest fitness values to act as parents. Common methods include roulette wheel, tournament, and ranking selection.

Crossover: Crossover generates offspring by exchanging genetic material between parent chromosomes. This can be done using one-point [6], two-point, or uniform crossover.

Mutation: Following crossover, some offspring undergo mutation, where genes are randomly altered with a low probability. This maintains genetic diversity [7]. This prototype will use Growth and Swap Mutation to expand tactical diversity.

Termination: The process stops when the algorithm no longer produces significant improvements or a maximum number of generations is reached, as noted in [8].

3. RESULTS AND FINDINGS

This paper identified the key components required to build the proposed optimization system. These components are the relevant performance benchmarks (KPIs) for the fitness function and the proven applications of GA in similar domains. Key Performance Indicators (KPIs) Strategy optimization refers to the process of improving strategy decisions to guarantee the most effective outcomes. The effectiveness of a strategy is evaluated using KPIs. According to a study by [9], several performance indicators (PIs) were identified as strong factors for winning and losing teams in both Rugby Union (RU) and Rugby Sevens (RS). These indicators, summarized in Table 1 and 2, show that winning teams score more tries, win more restarts, and (in RU) commit fewer penalties and miss fewer tackles. Another study [10] also identified technical determinants of tackle and ruck performance that relate to possession. These identified indicators for determining which aspects of gameplay should be prioritized in the fitness function.

Table 1. Key Performance Indicators (PIs) Related to Winning in RS

Performance Indicator	Significance (p value)	Effect Size (Cohen's d)	General Benchmark for Winning
Restarts received	$p < 0.001$	$d = 1.56$	Winning teams receive more restarts
Restarts won	$p = 0.001$	$d = 1.11$	Winning teams win more restarts
Line-outs won	$p = 0.013$	$d = 0.83$	Winning teams gain more possession
Tries scored	$p < 0.001$	$d = 2.13$	More tries scored

Table 2. Key Performance Indicators (PIs) Related to Winning in RU

Performance Indicator	Significance (p value)	Effect Size (Cohen's d)	General Benchmark for Winning
Restarts received	$p < 0.001$	$d = 1.18$	Winning teams receive more restarts
Restarts won	$p = 0.001$	$d = 1.21$	Winning teams win more restarts
Line-outs won	$p = 0.048$	$d = 0.65$	Winning teams win more line-outs
Tries scored	$p < 0.001$	$d = 1.43$	More tries scored

The literature review confirmed GA's effectiveness in solving complex optimization problems. GA has been successfully used to optimize training plans and game strategies in basketball [2], solve complex course and exam scheduling problems [11,12], and optimize team selection in fantasy cricket [13]. The review also found GA applied to diverse fields such as construction scheduling [14] and service scheduling in smart tourism [15]. These studies demonstrate GA's flexibility and its ability to produce optimal or near-optimal solutions quickly, even when handling difficult constraints. The success of GA in other sports [3,13] provides a strong foundation and highlights its potential for optimizing rugby strategy.

4. CONCLUSION

This research confirms that Genetic Algorithms (GAs) are a powerful and suitable method for solving complex optimization problems in various domains, including sports. The findings show that GA can successfully optimize strategies, training, and performance, providing a strong justification for its use in this project to optimize rugby strategies. This system can help coaches to organize match strategies more effectively and systematically. Furthermore, this research aims to fill a significant research gap especially in Malaysia. It is hoped that this data-driven approach will contribute to the development of a more advanced national sport.

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SENTIMENT CLASSIFICATION OF AIRLINE REVIEWS USING MACHINE LEARNING APPROACH

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ABSTRACT This study explores the use of machine learning (ML) methods for the sentiment analysis of customer reviews about Malaysia Airlines. The core problem is the necessity to correctly identify the polarity (positivity, negativity, or neutrality) of sentiments in unstructured online reviews, especially considering language nuances like sarcasm and the insufficient adaptability of prior models to the local, multilingual context of Malaysia. The main message is to identify the most efficient among four supervised ML models Support Vector Machine (SVM), Logistic Regression, Naïve Bayes, and Random Forest (RF) to classify sentiments. Core aims include identifying features contributing to sentiment polarity, developing and training classifiers using TF-IDF and LDA-based feature extraction, and assessing performance using accuracy, recall, precision, and F1-score. The expectation is to find an optimal model for optimized sentiment analysis to provide structured insights for airline operators. The study is limited to English-only text, excludes multimedia data, and uses a moderate scale of data

Keywords: Sentiment Classification; Airline Reviews; Machine Learning; LDA; Performance

1. INTRODUCTION

Airline customer reviews, widely shared online, influence potential travelers and reflect public perceptions of service quality. However, these reviews are often unstructured, diverse, and high in volume, making manual analysis difficult. Effective pre-processing and feature extraction such as using Latent Dirichlet Allocation (LDA) to uncover hidden topics help convert messy text into meaningful input for ML models. With increasing competition in Malaysia's aviation industry, feedback from platforms like Skytrax provides valuable insights into passenger experiences. This study focuses on analyzing Malaysia Airlines reviews, recognizing that model performance depends heavily on the quality and structure of the text data used.

The main problem lies in accurately identifying sentiment polarity positive, negative, or neutral in reviews that may contain sarcasm, blended emotions, or culturally specific expressions. Traditional ML models like Naïve Bayes, SVM, and Logistic Regression often struggle with these nuances because they rely too heavily on individual words rather than context. More advanced techniques, including n-grams, topic modelling (LDA), and word embeddings (Word2Vec), are needed to better understand relationships between words and improve classification accuracy. Although many ML algorithms have been used for sentiment analysis, there is no clear evidence on which model performs best for Malaysian airline reviews. Existing studies often use general or international datasets that do not reflect local language patterns or cultural behavior. As a result, global findings may not be reliable when applied in Malaysia. This gap highlights the need to evaluate different ML models specifically for Malaysian airline data to determine which algorithm provides the most accurate and interpretable sentiment classification, supporting Malaysia Airlines' customer satisfaction strategies. The main objective of this study is to determine the most effective sentiment analysis approach for Malaysia Airlines reviews, specifically by evaluating the performance of different machine learning algorithms in classifying sentiment.

2. MATERIALS AND METHODS

This study employs experimental research design with both qualitative and quantitative results. The quantitative results will include the accuracy, precision, recall, and F1-score of the methods. The experimental process is structured using the CRISP-DM (Cross-Industry Standard Process for Data Mining) method.

2.1. Data Preprocessing

The dataset consists of Malaysia Airlines passenger reviews taken from the Skytrax website for the year 2025. These reviews cover three categories: service, lounge, and seat. They were collected using Python's BeautifulSoup to extract text and rating information from the website. To prepare the data for supervised learning, the numerical ratings (1–10) were converted into sentiment labels (positive, neutral, or negative) based on fixed thresholds. Following the method in [2], ratings of 1–3 are classified as negative, 4–6 as neutral, and 7–10 as positive, as shown in Table 2.

Table 2. Performance result using LDA

Rating	Polarity
1-3	Negative
4-6	Neutral
7-10	Positive

Before training the models, the text reviews undergo several pre-processing steps to improve data quality. First, sentiment segmentation splits each review into individual sentences so mixed opinions within long reviews can be analyzed more accurately. Next, tokenization using regular expressions breaks sentences into word tokens, which is a key step in NLP and works well for informal online language. Then, punctuation and noise removal is performed to clean elements like usernames, URLs, numbers, and unnecessary symbols, while keeping hashtag words for their potential meaning. After that, the text is converted to lowercase and stop words are removed using a predefined list. Finally, lemmatization reduces words to their base form to standardize vocabulary, reduce dimensionality, and improve the model's overall performance.

2.2. Feature Extraction Technique

The most suitable feature extraction method for this study is TF-IDF and Latent Dirichlet Allocation (LDA). LDA is used to expose latent topics (e.g., flight delay, customer care, seating comfort) by treating reviews as a mixture of topics. This approach improves sentiment classification by focusing on the sentiment related to specific aspects of the services, rather than the review in totality.

2.3. Supervised Machine Learning Methods

The dataset will be split into 80% for training and 20% for testing. This study applies four supervised machine learning algorithms Support Vector Machine (SVM), Logistic Regression, Naïve Bayes, and Random Forest using their default parameter settings to classify reviews into sentiment categories. Using default parameters is a standard practice for establishing baseline results before tuning, and several recent studies confirm the effectiveness of these algorithms in text classification tasks with such setups [7]. Model performance is then rigorously assessed through evaluation techniques to ensure generalizability and reliability. SVM classification is evaluated using a confusion matrix and 5-fold cross-validation [8]. Then, Naïve Bayes is evaluated via standard tests and evaluation of the model [9]. Random Forest uses K-Fold Cross Validation [10].

2.4. Evaluation of Metrics for Classifier Performance

Model performance will be evaluated using four main metrics from the confusion matrix: Accuracy, Precision, Recall, and F1-score. Accuracy shows how many sentiment predictions (positive, neutral, or negative) are correct overall. It is calculated by dividing the total correct predictions (TP + TN) by all predictions (TP + TN + FP + FN). Precision measures how many of the reviews predicted as positive are positive, using the formula $TP / (TP + FP)$. Recall shows how well the model can detect actual positive reviews by calculating $TP / (TP + FN)$. This is important for identifying all satisfied customers. The F1-score combines precision and recall into one value using their harmonic meaning, making it

useful when the dataset is imbalanced. According to [6], these metrics differ in properties such as class sensitivity and chance correction, which helps determine which metric is most appropriate for different classification tasks.

3. RESULTS AND FINDINGS

The performance of the four supervised machine learning models Logistic Regression, Random Forest, Naïve Bayes, and SVM was evaluated using two different feature extraction techniques: Latent Dirichlet Allocation (LDA) and Term Frequency-Inverse Document Frequency (TF-IDF). The performance metrics assessed were Accuracy, F1-score, Recall, and Precision. The results of using LDA as the feature extraction technique are presented below:

Table 2. Performance result using LDA

Machine Learning	Accuracy	F1	Recall	Precision
Logistic Regression	0.500	0.491	0.500	0.506
Random Forest	0.506	0.472	0.506	0.487
Naïve Bayes	0.443	0.443	0.443	0.482
SVM	0.509	0.509	0.509	0.521

The Support Vector Machine (SVM) model demonstrated the best performance across all metrics when using LDA features, achieving the highest Accuracy (0.509), F1-score (0.509), Recall (0.509), and Precision (0.521). Naïve Bayes recorded the lowest performance.

The results using TF-IDF as the feature extraction technique are presented below:

Table 3. Performance result using TF-IDF

Machine Learning	Accuracy	F1	Recall	Precision
Logistic Regression	0.506	0.477	0.506	0.493
Random Forest	0.521	0.460	0.521	0.508
Naïve Bayes	0.509	0.452	0.509	0.493
SVM	0.515	0.456	0.515	0.498

When utilizing TF-IDF, the Random Forest (RF) model outperformed the others in terms of Accuracy (0.521) and Recall (0.521). However, the F1-score and Precision were marginally lower than those achieved by SVM using the LDA feature. Overall, the performance metrics for all models when using TF-IDF were slightly higher in Accuracy compared to the LDA approach. It shows that these approaches struggle with the complex, nuanced structure of the content. Significant improvements are required for the next experiment: future research should use other datasets to better capture local language patterns and go beyond frequency-based feature extraction (such as TF-IDF and LDA) to investigate more complex, context-aware methods. In particular, since standard models just offer a baseline, the following stage should look at deep learning models (such CNNs or LSTMs) to get far higher classification scores and improved generalizability.

4. CONCLUSION

This study aimed to identify the most effective machine learning approach for sentiment classification of Malaysia Airlines reviews, utilizing four supervised models and two feature extraction techniques (LDA and TF-IDF). The Random Forest (RF) classifier combined with the TF-IDF feature extraction technique achieved the highest overall Accuracy (0.521) and Recall (0.521), demonstrating its slight superiority in predicting sentiment polarity on this dataset. Conversely, the Support Vector Machine (SVM) model using LDA features delivered the most balanced performance, achieving the highest F1-score (0.509) and Precision (0.521) for that specific feature set. This finding aligns with literature suggesting that ensemble methods like RF are often robust for noisy, unstructured text like reviews, while SVM excels in high-dimensional spaces like LDA topic distributions. The observed performance

metrics, clustered around the 50% mark, suggest that while traditional ML models provide a baseline, the complex nature of the data, especially the presence of nuanced language and potential class imbalance, indicates that further refinement in feature engineering or the exploration of deep learning models would be crucial for achieving substantially higher classification scores and greater generalizability. Future work should focus on multilingual datasets and deep learning approaches to enhance performance.

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ATIQAHSNART: IOT SPEECH BASED HOME ASSISTANT

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ABSTRACT The *AtiqahSmart* is a smart home assistant that enables users to control home devices through voice commands and retrieve weather updates. The mobile Android system offers secure user access, persistent storage of device states and retrieval of real-time news. The smart home assistant aims to optimize functionality, firstly, with seamless integration of a refined BERT-based natural language processing (NLP) model and IoT for command classification. This hybrid approach allows the assistant to accurately distinguish between different command types, specifically used for handling the device-specific control actions and context-based weather queries. Secondly, rule-based logic is responsible for processing parameters like time, location and device settings (brightness, speed or temperature) used for personalized context. The rule-based component ensures that responses are contextually relevant and actionable with an integrated ESP32 microcontroller to demonstrate device control. The mobile system supports both voice and text and can easily incorporate additional devices or services with minimal reconfiguration. The key advantage of this solution is its cost-effective design and adaptability, feasible adoption across different user groups and environments. In conclusion, *AtiqahSmart* is a cost-effective AI-powered personalize and user-friendly system with full potential to be commercialized as a viable smart home assistant for everyday decision-making and automation.

Keywords: Smart home, BERT, NLP, IoT

1. INTRODUCTION

Smart home assistants have become an increasingly important part of modern living, offering users convenience in managing daily routines. By automating tasks and enabling hands-free interaction [1], these systems can significantly improve accessibility particularly for elderly users and those with physical limitations [2]. However, prior research has explored that many existing assistants require web-based integration and manual configuration for IoT devices which limiting their usability and deployment flexibility [3].

Unfriendly and not seamless assistants often limit the interaction between users and the system, resulting in responses and actions that feel rigid and less intuitive [4]. By unfriendly system, it is meant that many assistants require very rigid phrasing and offer little personalization. If users don't say the exact words, the system may not respond correctly. On the other hand, users face challenges in giving natural commands and managing devices smoothly by not being seamless which makes the experience less convenient and more frustrating [5]. This creates barriers for new adopters and reduces the effectiveness of speech-based technologies.

Therefore, *AtiqahSmart* is a mobile application designed to address this need by providing a voice-driven solution for smart home management. The application focuses primarily on IoT device control which allowing users to switch devices on or off, adjust settings such as brightness, fan speed or temperature and manage multiple rooms seamlessly. In addition, it extends its functionality by delivering real-time weather updates that supporting everyday decision-making with contextual information.

2. MATERIALS AND METHODS

The system was developed as a mobile application integrating speech recognition, command classification and IoT control as in Figure 1 that illustrates the overall architecture. User input is captured either through speech or text and processed within the application. Then, a fine-tuned BERT model classifies the command into one of four personalization categories which identifying whether the query is device-based or context-based. Following classification, a rule-based engine extracts actionable

parameters such as device name, room, action type, speed, brightness, time or weather location.

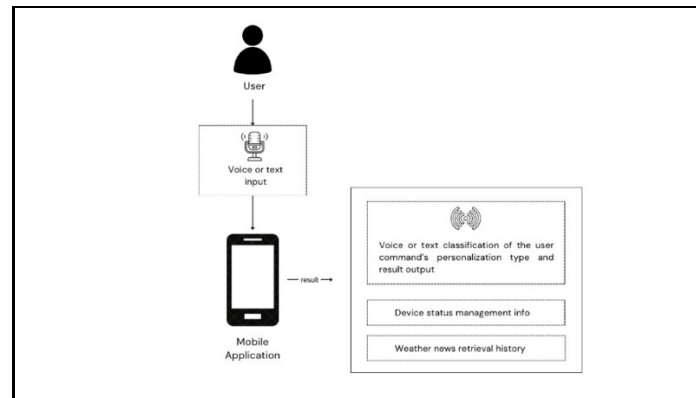


Figure 1. System Architecture of AtiqahSmart

The resulting instruction is then mapped to the appropriate output either an executed IoT device command or a real-time weather update retrieved via API. Finally, device states and weather history are stored for reference within the application. This hybrid design ensures that natural language input is interpreted accurately while maintaining deterministic control behavior. The architecture supports multi-room device management and can be extended to additional smart appliances without retraining the model.

3. RESULTS AND FINDINGS

To validate the design of AtiqahSmart, a BERT classification model as in Figure 2 was trained to classify user commands into four personalization categories. Among these, two key categories of context-based weather queries and device-based control are directly supported by rule-based modules.

```

=== Final Evaluation ===
Accuracy: 0.914572864321608
Classification Report:

```

	precision	recall	f1-score	support
content-based	1.00	0.73	0.84	22
context-based	0.91	1.00	0.95	80
device/environment-based	0.98	0.88	0.92	49
non-personalized	0.84	0.90	0.87	48
accuracy			0.91	199
macro avg	0.93	0.88	0.90	199
weighted avg	0.92	0.91	0.91	199

Figure 2. Evaluation of Trained BERT Classification Model

The BERT classification model enables the system to distinguish between:

- Device-based commands (e.g., “Turn on the bedroom fan”)
- Context-based commands (e.g., “What’s the weather in Shah Alam tomorrow?”)
- Other unsupported types, which are flagged accordingly

Table 1 Device Command Parsing and Firebase Update Results

Command Example	Extracted Room	Extracted Device	Action	Extra Setting	Result (Firebase Update)
Set the bedroom fan to high speed	Bedroom	Fan	On	Speed: High	/rooms/bedroom/fan/state = "on", speed = "high"
Switch off the	Living	Air	Off	None	/rooms/living room/ac = "off"

living room air conditioner	Room	Conditioner			
Turn on the kitchen light	Kitchen	Light	On	None	/rooms/kitchen /devices/light = "on"

The results on Table 1 shows some examples on device command parsing where rule-based logic is used then interprets essential factors such as:

- Time – *now, tomorrow, at 9PM*
- Location – *home location from Firebase or mentioned in the query*
- Device type – *fan, light, air-conditioner*
- Action – *turn on, turn off, adjust, switch, set*
- Extra settings – *brightness, speed, temperature*

This hybrid approach ensures that each classified command produces a clear, accurate and actionable response. In short, BERT handles classification while the rule-based logic makes the command executable.

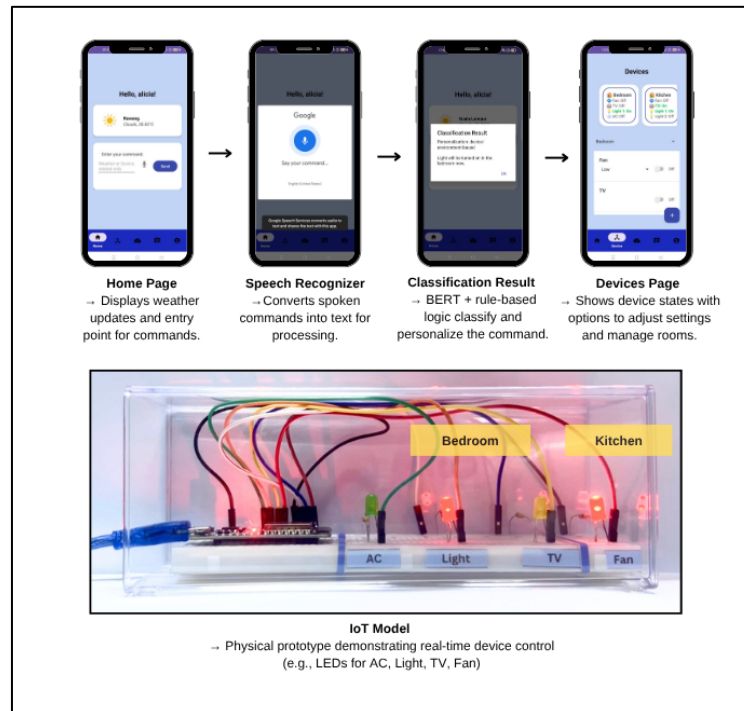


Figure 3. Mobile Application UI and IoT Model Snapshots

Based on Figure 3, the application also has been tested and demonstrated with real-world scenarios to showcase its capabilities that aligned with the project's objective:

- **Voice-based Device Control:** For example, the command *"Turn on the bedroom light"* is classified as a device-based action. Rule-based logic identifies the device and room then the command is executed instantly. The result is reflected in real time both in the app and through a connected IoT device (LED indicator).
- **View and Adjusting Device Settings:** Users can view summary of device states across all rooms and also manually fine-tune IoT device control such as setting a fan to high speed or dimming a light. These commands are recognized and directly synchronized with Firebase, ensuring that device states are updated accurately.

- Adding Devices and Rooms: The system supports scalability by enabling new rooms and IoT devices to be added easily. For instance, adding a *Dining Room* with a *Light* device configured with brightness settings demonstrates how the platform adapts to user growth.
- Weather News Retrieval: Context-based queries such as “*What’s the weather in Shah Alam tomorrow?*” are supported. The assistant retrieves real-time updates from the OpenWeatherMap API, considering both the user’s home location (from Firebase) and any location specified in the query.

4. CONCLUSION

In conclusion, *AtiqahSmart* demonstrates how AI-powered classification, rule-based logic and IoT integration can be combined into a cost-effective and user-friendly smart home assistant. By focusing on speech-based IoT device control and contextual weather retrieval, the system delivers a hands-free, accessible and personalized experience. It addresses key gaps in existing assistants which improving user-friendliness, seamlessness and adaptability while showcasing potential for real-world commercialization.

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RESPONSIBLE AI ADOPTION FOR HYPER-PERSONALISATION VIA LEARNING ANALYTICS IN MALAYSIAN HIGHER EDUCATION: INSIGHTS FROM THE SINAR PROTOTYPE

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ABSTRACT- Responsible AI (RAI) is essential for the safe and trustworthy deployment of AI systems in education, where fairness, transparency, and human oversight directly influence student outcomes. However, Malaysian higher education lacks operational models for translating RAI principles into practice, especially for learning analytics systems that must adapt to diverse data sources and academic workflows. Early upscaling of the SINAR, a prototype system for hyper-personalisation via learning analytics (comprising integration of predictive modelling, digital-twin trajectories, and interpretability) exposed key challenges including inconsistent data structures, potential fairness risks, unstable explanations, misaligned workflows, and the absence of institutional governance mechanisms such as drift monitoring or model documentation. These challenges intensified when extending the system across multiple courses, signalling a need for MLOps processes to support continuous monitoring and controlled model evolution. This paper presents a Malaysian-aligned RAI adoption checklist for SINAR and maps the IMPEX monitoring and model-editing prototype into the RAI–MLOps lifecycle, offering a practical pathway for embedding ethical and operational safeguards into future AI deployments.

Keywords: Responsible AI; Learning Analytics; Educational Data Mining; Hyper-Personalised Student Support

1. INTRODUCTION

Artificial intelligence (AI) is increasingly embedded in higher education through learning analytics (LA) and educational data mining, enabling early detection of at-risk learners and personalised intervention pathways. Early systems such as Purdue’s Course Signals [1] demonstrated the potential of predictive analytics, while recent advances in digital-twin modelling provide richer, temporal representations of student trajectories [2]. Explainable AI (XAI) research further emphasises the need for transparent and interpretable insights to support educator trust and appropriate use [3].

In Malaysia, national initiatives including the National AI Roadmap 2021–2025 [4], MyDIGITAL Blueprint [5], and Digital First@MOHE [6] underscore the transition toward AI-enabled, data-driven education, complemented by governance frameworks such as AIGE [7] and the National AI Office principles [8] that highlight fairness, accountability, explainability, and human oversight.

Despite these developments, a significant translational gap remains between AI research prototypes and responsible institutional deployment. Data ecosystems are fragmented and inconsistent, educator readiness for interpreting AI insights is uneven, and fairness assessments are rarely conducted. Crucially, institutional governance structures such as AI oversight committees, documentation protocols, and drift-monitoring mechanisms are largely absent.

We argue that these challenges cannot be addressed without explicit RAI safeguards. We present lessons learned from early upscaling and deployment of SINAR (Supportive Insights for Nurturing Academic Resilience), a learning analytics-driven prototype that integrates predictive analytics, digital-twin trajectories, resilience indicators, and interpretability features. Accordingly, this paper contributes (i) a Malaysian-aligned RAI checklist to guide institutional adoption of SINAR, and (ii) a mapping of the IMPEX prototype [9] (an AI monitoring and model-editing tool within SINAR), into the MLOps and RAI lifecycle, demonstrating how technical infrastructure can operationalise ethical and governance requirements.

2. LESSONS LEARNED ON RESPONSIBLE AI IMPLEMENTATION FROM EARLY WORKS FOR ADOPTION OF SINAR

Building on prior works in explainable interactive machine learning [9], LSTM-based prediction [10], and Malaysian educator acceptance studies [11], the expansion of SINAR uncovered new requirements for model editing, explanation stability, counterfactual reasoning, and continuous monitoring across multiple courses and faculties; conditions that shift the system into an MLOps environment.

2.1 Responsible AI Framework for SINAR's Implementation

These challenges resonate strongly with Malaysia's National Guidelines on AI Governance and Ethics (AIGE), which emphasise fairness, explainability, robustness, human-centred values, and accountability as mandatory requirements for AI systems [7]. Similar expectations are outlined in the National AI Office's Governance and Ethics Principles, which underscore the need for risk assessment, oversight mechanisms, and responsible system management [8]. At the international level, the OECD AI Principles reinforce transparency, security, human oversight, and inclusive growth as prerequisites for trustworthy AI [13]. Together, these Malaysian and global frameworks highlight the necessity of treating SINAR's upscaling as both a technological and a governance endeavour.

Synthesising these insights, six RAI dimensions (Figure 1) emerge as essential checklist for guiding SINAR's institutional adoption: (1) clarity of purpose and intended use, establishing transparency about what SINAR is, and is not designed to do; (2) data and model quality assurance, ensuring interoperability, completeness, continuous validation, and drift detection; (3) fairness and inclusivity, supported by routine bias auditing and subgroup analysis; (4) transparency and explainability, delivered through structured model documentation, narrative reasoning, and uncertainty visualisation; (5) governance and accountability, anchored in oversight committees, audit processes, and documented decision pathways; and (6) human oversight and capacity building, ensuring educators remain central decision-makers and receive adequate training to interpret and act on AI outputs.

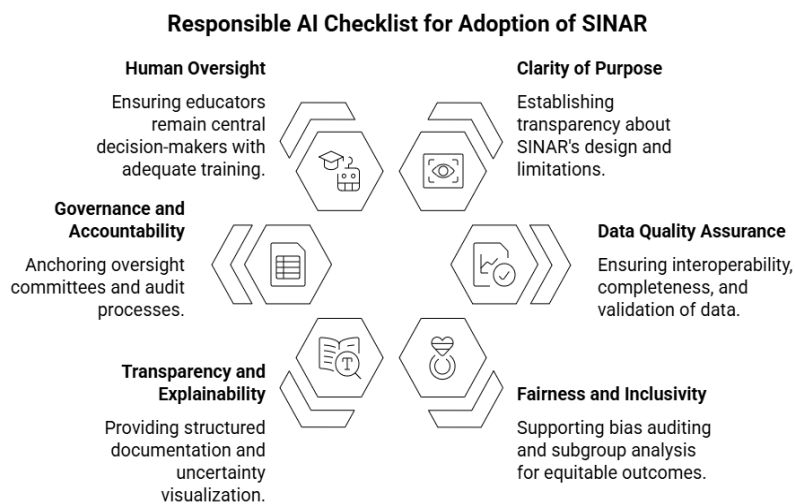


Figure 1. Responsible AI Checklist for Adoption of SINAR

2.2 Expansion of SINAR's Deployment through MLOps

In preparing to upscale the prototype into a multi-course, institution-wide system, it became evident that RAI cannot be meaningfully implemented without strong MLOps foundations. MLOps is the practice of managing machine learning systems so they can run reliably in the real world. It covers everything needed after a model is built encompassing cleaning and updating data, tracking versions, checking for errors or changes over time, and making sure the model stays accurate and safe.

Just as DevOps ensures software runs smoothly, MLOps ensures AI models continue to work properly as data, courses, or users change. In education settings, MLOps is crucial because it helps keep AI predictions transparent, up-to-date, and trustworthy for students and educators. This links directly to RAI principles, which are fairness, transparency, accountability, and human oversight; cannot be achieved without the monitoring, documentation, and control processes that MLOps provides. In education settings, combining MLOps with RAI helps ensure AI predictions remain transparent, up-to-date, and trustworthy for students and educators.

Scaling the model across different courses introduces variability in data quality, assessment structures, cohort characteristics, and feature semantics conditions that require systematic data pipelines, monitoring mechanisms, experiment reproducibility, version control, and model documentation. These MLOps components form the operational backbone that enables core RAI expectations such as fairness auditing, explanation stability, performance drift detection, transparent model evolution, and traceable human oversight. In this context, MLOps ensures that RAI principles are not treated as abstract ethical aspirations but are embedded directly into the technical lifecycle of the system. Thus, RAI provides the normative guidance for what the system must safeguard, while MLOps provides the infrastructure through which those safeguards can be reliably implemented at scale.

The IMPEX module [9] with SINAR prototype directly operationalises the intersection of RAI and MLOps (Figure 2). While RAI defines what the system must protect (fairness, transparency, human oversight, and accountability), IMPEX provides the mechanisms that make these principles technically achievable. Its adaptive counterfactual generator, interactive surrogate editing interface, drift-monitoring tools, and explanation-validation workflows function as MLOps components that maintain model reliability and transparency over time. IMPEX enables controlled model evolution, logs human edits for accountability, surfaces drift signals for timely retraining, and supports explanation checks to ensure stability and truthfulness. In doing so, it transforms RAI from a set of high-level ethical commitments into concrete, operational practices. Thus, IMPEX is not merely a decision-support prototype, it is the technical backbone through which RAI governance becomes executable, auditable, and scalable within a real institutional AI lifecycle.

Responsible AI Deployment with SINAR				
Characteristic	Responsible AI (RAI)	MLOps	IMPEX module in SINAR	Trustworthy & Scalable AI
Focus	Fairness, Transparency, Accountability, Oversight	Data Pipelines, Monitoring, Versioning, Governance, Documentation, Drift Checks, Audit Trails	Adaptive Counterfactual Generation, Interactive Surrogate Editing, Explanation Validation, Performance & Drift Monitoring, Edit Logs & Model Evolution Tracking, Uncertainty & Bias Signals	Fair & stable predictions, Transparent explanations, Human oversight maintained, Model behaviour traceable & auditable, Safe scaling
Purpose	RAI Requirements	MLOps Implementation Layer	Execution of RAI in Practice	Trustworthy & Scalable AI Deployment

Figure 2. Responsible AI Deployment in MLOps through IMPEX

3. CONCLUSION

This paper used the SINAR prototype as a practical lens to understand what is required for implementing RAI in hyper-personalised learning analytics within Malaysian higher education. The feasibility analysis highlighted persistent challenges ranging from inconsistent data quality and potential fairness risks to limited interpretability, workflow misalignment, and the absence of structured AI governance. By mapping these issues against Malaysia's national AI governance guidelines and international

trustworthy-AI principles, the paper proposed a Malaysian-aligned RAI framework and governance checklist to support ethical, transparent, and accountable deployment of learning analytics systems. Future advances in SINAR will require multimodal digital-twin modelling, stronger prescriptive analytics, robust monitoring infrastructure, and pilot studies conducted under formal RAI governance to evaluate fairness, stability, and educator readiness. The lessons learned contribute toward building a responsible, equitable, and scalable foundation for AI-enabled hyper-personalisation in Malaysian higher education.

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REVIEW ON FINANCIAL INFORMATION RETRIEVAL: A LITERATURE REVIEW

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ABSTRACT The rapid growth of digital financial data - from news articles, analyst reports, regulatory filings, and social media has created a need for advanced Information Retrieval (IR) systems in the financial sector. Financial Information Retrieval (FIR) aims to efficiently index, retrieve, and rank relevant financial documents to support analysis, risk assessment, and market prediction. Over time, financial information retrieval has evolved from simple keyword-based searches to advanced methods integrating machine learning and natural language processing (NLP). These approaches enable systems to interpret sentiment, classify financial documents, and detect relevant market events with greater accuracy. Techniques such as information extraction, text classification, and transformer-based models have significantly improved event detection, sentiment scoring, and decision support. Moreover, domain-specific models like FinBERT demonstrate superior performance by capturing nuanced financial language and sentiment more effectively than general-purpose language models. Despite these advancements, challenges remain in integrating heterogeneous data, improving model explainability, and combining textual and numerical financial signals. To address these gaps, this research proposes a hybrid FIR framework that integrates textual financial documents with numerical time-series indicators. This approach aims to strengthen predictive capabilities and provide a more holistic understanding of financial market behaviour, contributing to more accurate and interpretable financial analytics.

Keywords: Financial Information Retrieval (FIR); Information Retrieval (IR); Natural Language Processing (NLP); Text Classification; Financial Documents; Financial Analytics; FinBERT's

1. INTRODUCTION

Information Retrieval (IR) is finding material (usually documents) of an unstructured nature (usually text) that satisfies an information need from within large collections (usually stored on computers) [1]. It is a computational technique used by search engines and system to store, retrieve and rank documents from a large collection of text documents based on users queries. In an IR system, the first step must involve indexing, where documents within a collection are systematically organized into a structured representation known as an index [2]. This process captures and stores essential information from each document in a unified structure to enable faster and more accurate retrieval operations [2].

IR allows analyst, traders, portfolio managers and other financial professionals to speed up the need to access and analyse financial analysis, risk assessment, and market forecasting in an efficient method. Quantitative trading relies on IR to extract signals from unstructured data like news, social media, and earnings call transcripts [3]. Analysts scan text for patterns such as merger rumours or policy changes and search historical data for similar past events [3]. With the increasing volume of financial information generated from various sources, IR has proven to become an essential tool in the financial domain. Financial Information Retrieval (FIR) applications also now have to extend market sentiment retrieval, event-driven stock analysis, fraud detection, and automated report summarization.

1.1 Components of IR

Recent literature has demonstrated steady progress in applying modern IR techniques to financial data. Early studies focused on keyword-based retrieval for corporate disclosures and news filtering. As discussed by Gupta A., Dengre V., Kheruwala H. A., Shah M. [4], Information extraction (IE) is one of the methods used to extract

predefined data types from a text document and it aim for object identification by extracting relevant information from the fragments and then putting all the extracted pieces in a framework.

Natural language processing (NLP) is a subfield of Artificial Intelligence (AI) and Computational Linguistics that focuses on enabling computers to understand, interpret, and generate human language in a meaningful way. In the financial sector, it has been used to assess a firm's current and future performance, domain standards, and regulations. Lewis and Young [7] discussed the importance of text mining in financial reports and they preferred the NLP methods. They highlighted the exploding growth of unstructured textual data in corporate reporting, which opens numerous possibilities for financial applications. According to the authors, they highlight two generic benefits of applying NLP methods to analyse outputs from the corporate reporting process. The first generic benefit is the ability to process large volumes of content at relatively low cost, thereby helping to mitigate concerns about information overload as the quantum of disclosures increase to reflect preparers' wider social responsibilities. The second generic benefit is the ability of NLP methods to detect latent features in the data that even the closest manual analysis may struggle to identify.

Another component of Information Retrieval (IR) is Text classification, which is an automatic process of sorting documents into predefined categories to organize and manage large amounts of text data. It uses algorithms to learn patterns from text and make predictions, often leveraging techniques like Naive Bayes, SVMs, and transformer models. According to Sebastiani [8], text classification is one of the most important applications of machine learning in IR, as it allows systems to efficiently manage and retrieve relevant information. Brindha S [5] have compared the performance of various text classification techniques, namely NB, k-nearest neighbour (KNN), SVM, decision tree, and regression, and found that based on the precision, recall, and F1 measures, SVM provided better results than the others.

Later, IR have adopted machine learning and deep neural models, including BERT and its finance-specific variants such as FinBERT, to improve semantic understanding and ranking performance. BERT is conceptually simple yet powerful language representation model introduced by Devlin [9]. The term BERT stands for *Bidirectional Encoder Representations from Transformers*. Unlike recent language representation models, BERT is designed to pre-train deep bidirectional representations from unlabelled text by jointly conditioning on both left and right context in all layers. As a result, the pre-trained BERT model can be fine-tuned with just one additional output layer to create state-of-the-art models for a wide range of tasks, such as question answering and language inference, without substantial task-specific architecture modifications. While FinBERT, which is a language model based on BERT for financial NLP tasks was introduced by Araci, D. [10]. They have evaluate FinBERT on two financial sentiment analysis datasets and have achieve the state-of-the-art performance on FiQA sentiment scoring and Financial PhraseBank, by outperforming general-purpose language models. The model learns to classify sentences or documents into sentiment categories: of Positive, Negative or Neutral. By learning from financial terminology and tone (e.g., "bullish outlook," "missed earnings target"), FinBERT captures contextual meaning more accurately than general NLP models. These results highlight FinBERT's capability to capture the subtle nuances of financial language, making it highly effective for sentiment scoring and other finance-related NLP applications.

2. CONCLUSION

Information Retrieval (IR) in Finance has become a critical component in analysing and processing the growing volume of financial data. From its early reliance on keyword-based searches for corporate disclosures and market news, IR has progressed toward intelligent frameworks that integrate natural language processing (NLP), text classification, and deep learning models. These modern systems enable a deeper understanding of semantic meaning, context, and sentiment within unstructured financial text, significantly enhancing prediction, compliance, and decision-making capabilities. However, as existing research still faces challenges related to data integration, to address these gaps, our future work aims to develop a hybrid Financial Information Retrieval framework that integrates numerical time-series data with textual financial documents (e.g., news, sentiment and analyst reports).

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